

Strategies for building domain specific knowledge for single cell genomics

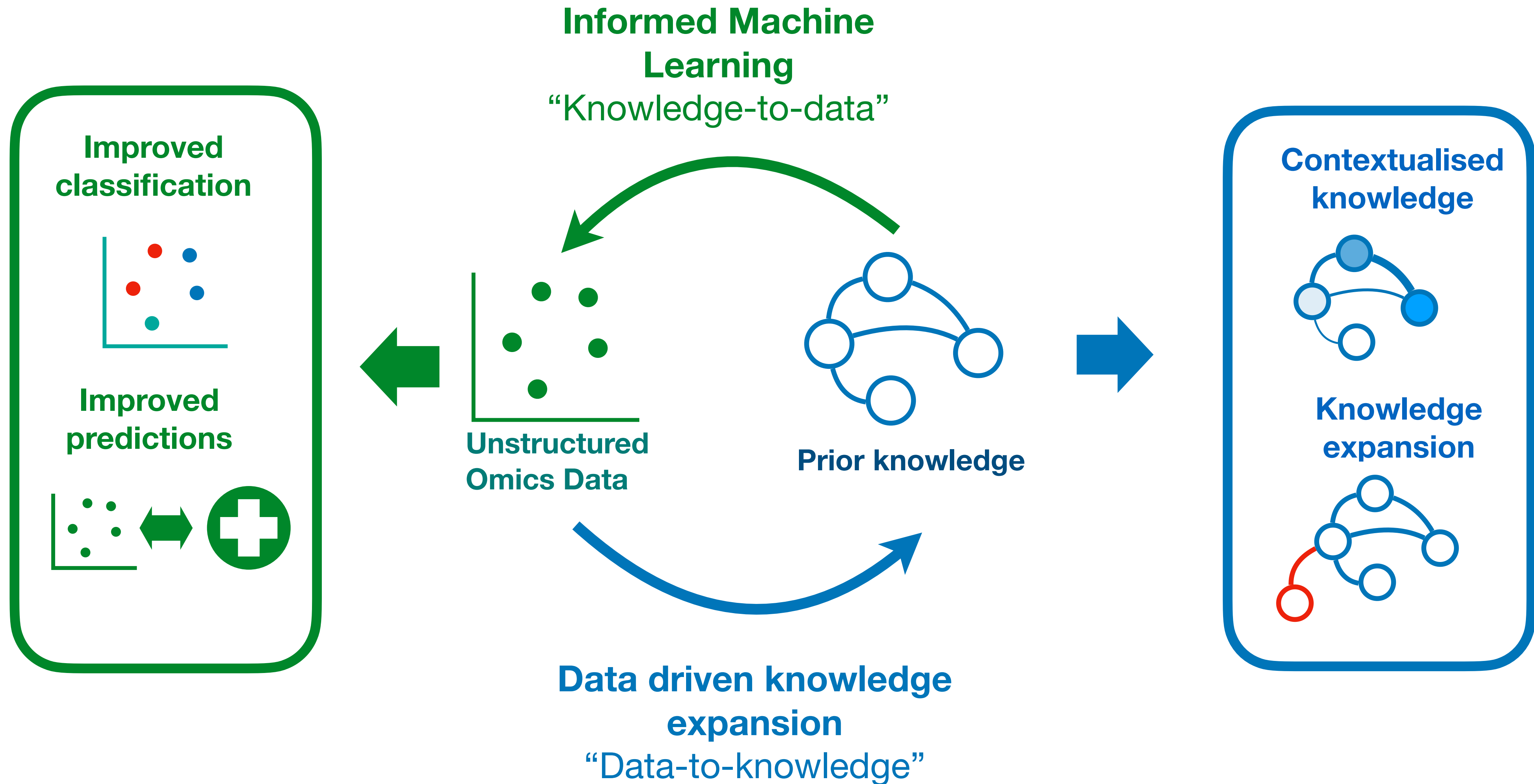
Carl Herrmann

BioCypher Workshop - 17.06.2026

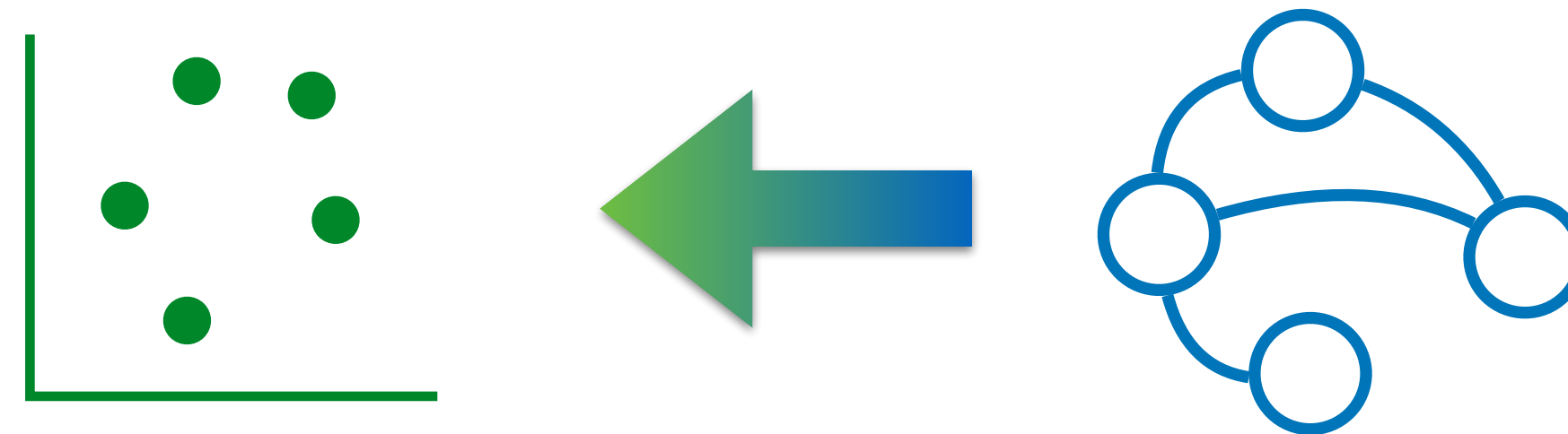


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Our vision

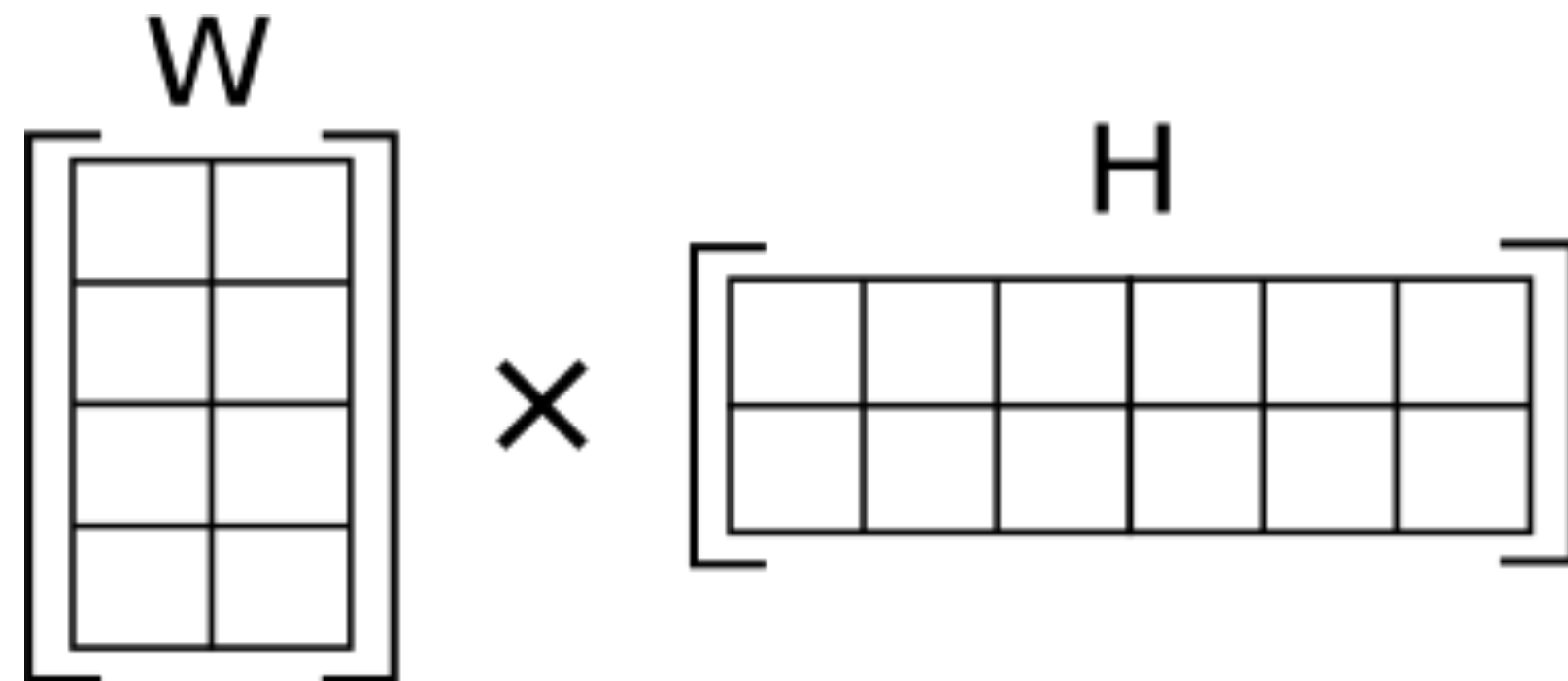


Injecting knowledge into data : “Knowledge to data”

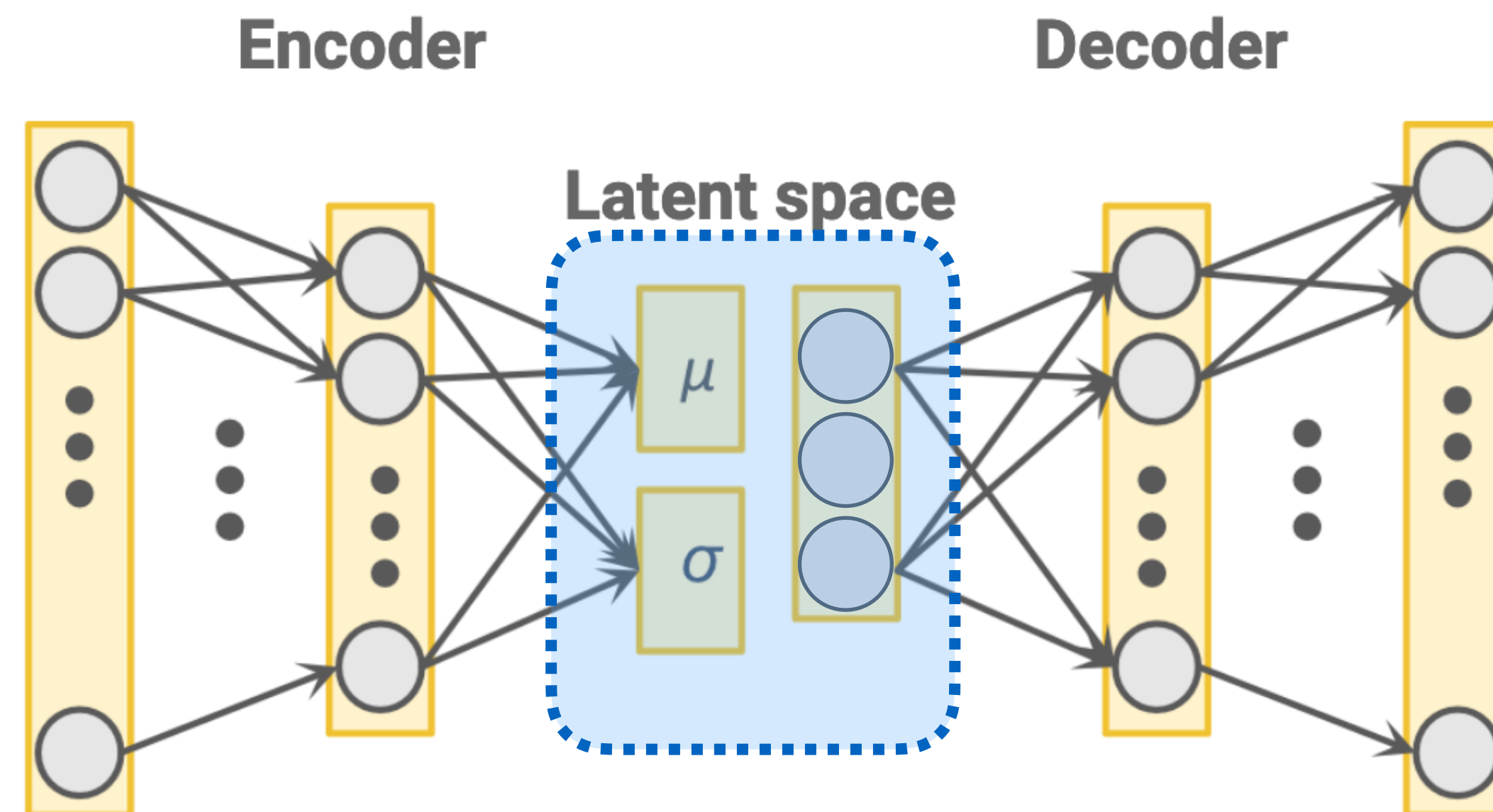


What models?

Non-negative matrix factorization

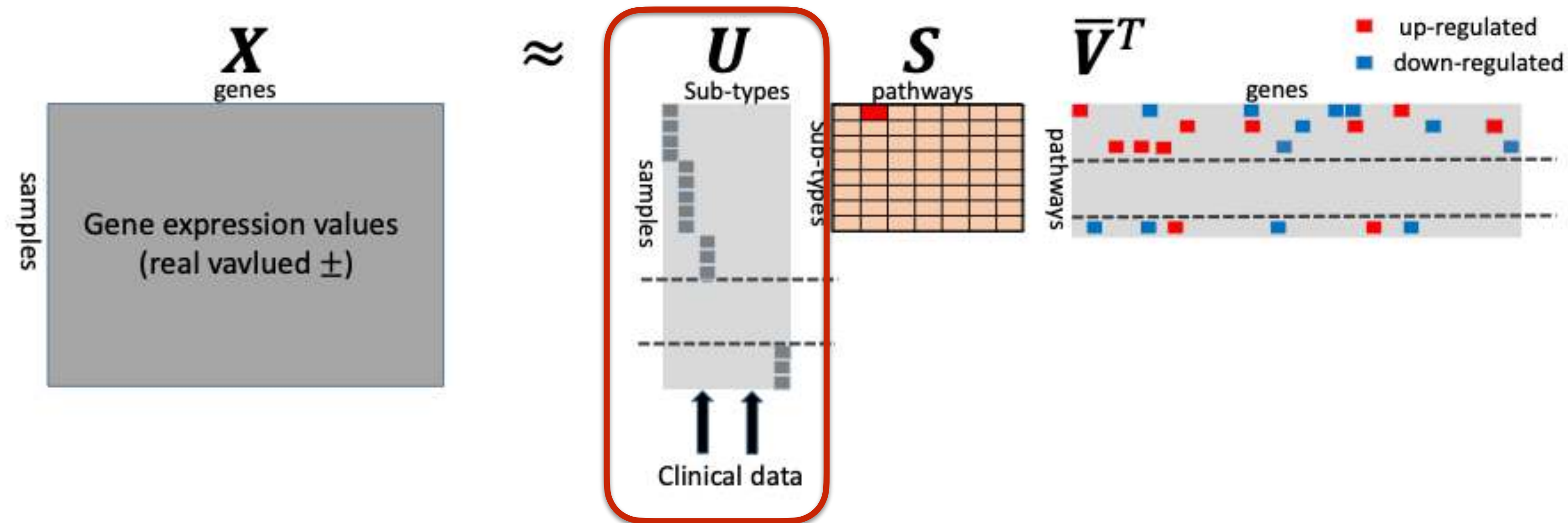


Variational autoencoder



Unsupervised models !

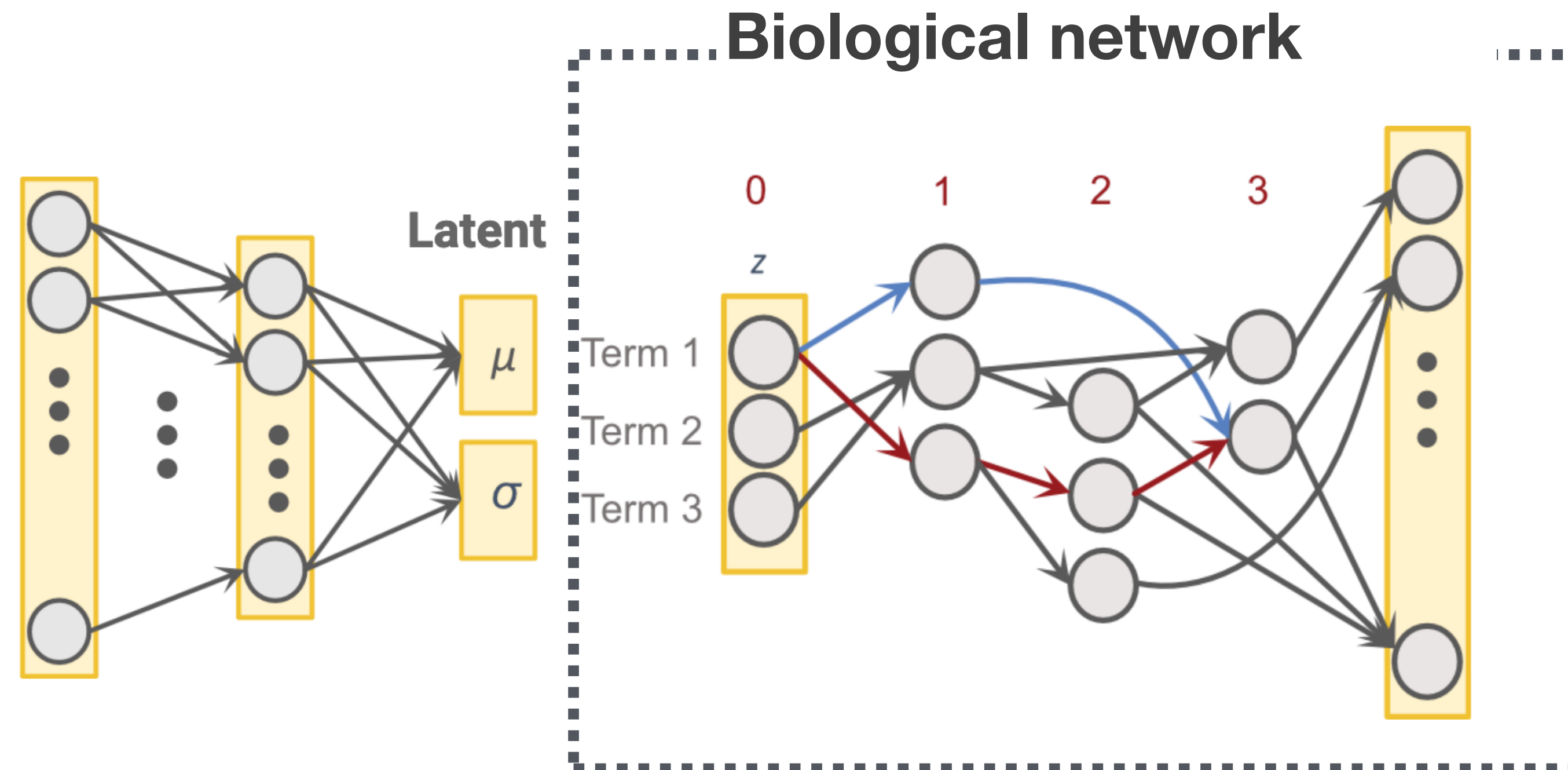
How can we inject knowledge into these models?



- Include prior clinical / pathway information into the tri-NMF
- Clinical information (U) will guide the decomposition (S, V)

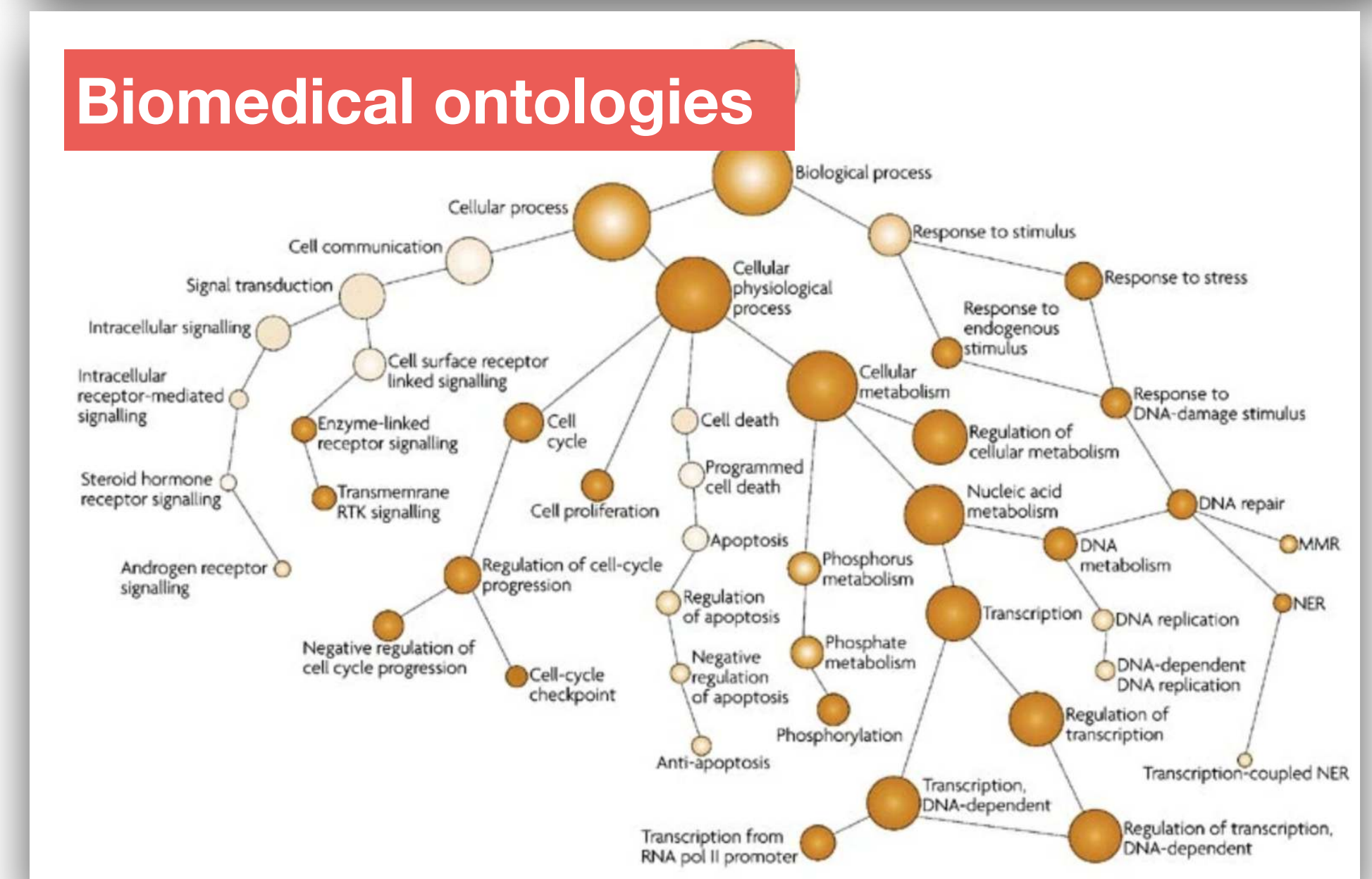
[Park et al., 2019]

Injecting knowledge into VAE

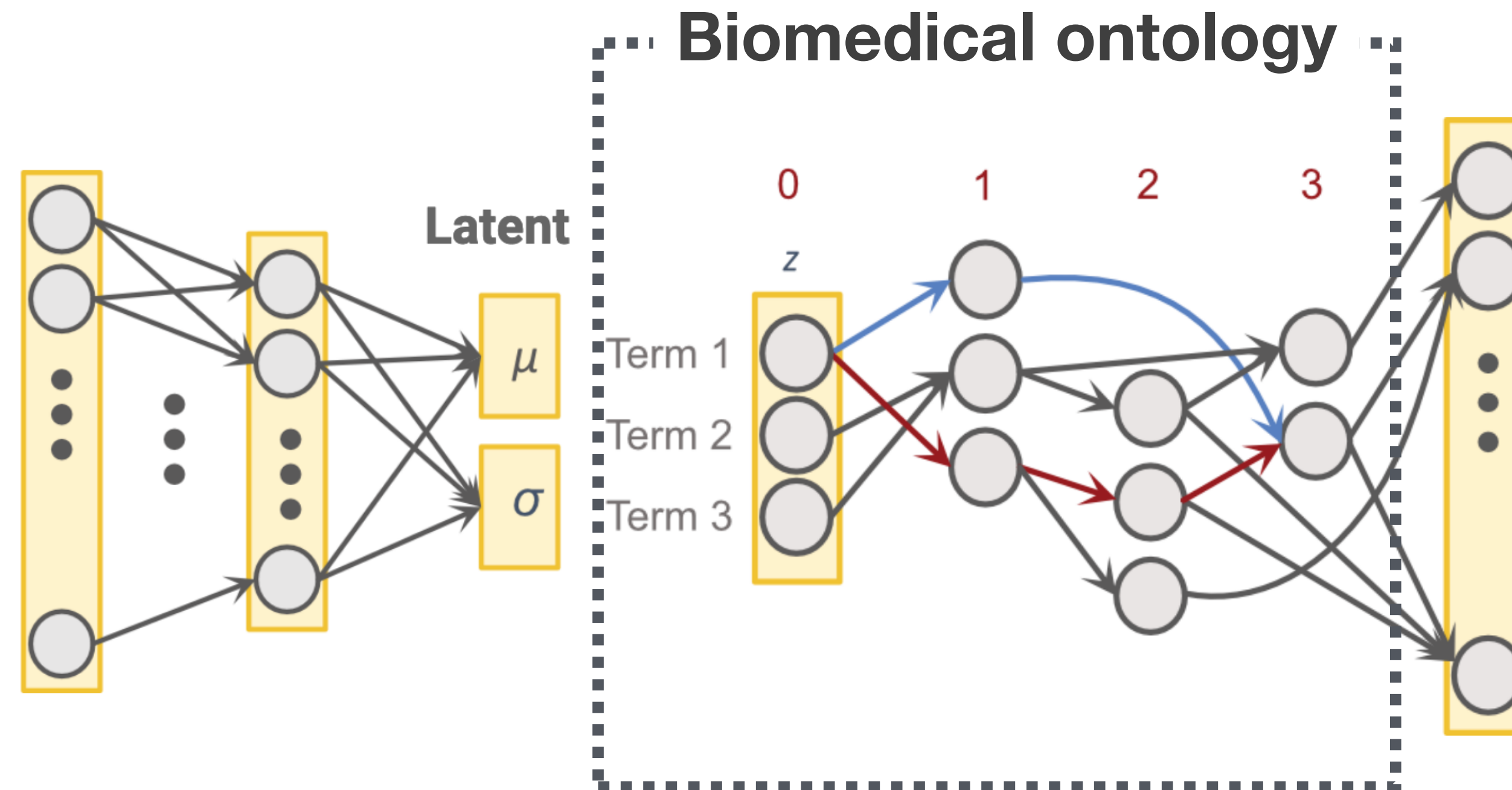


- Replace the decoder by a **biologically informed network**
- **Neuron activations** $\approx$ biological activity
- **Edge weights** represent the strength of the influence between nodes

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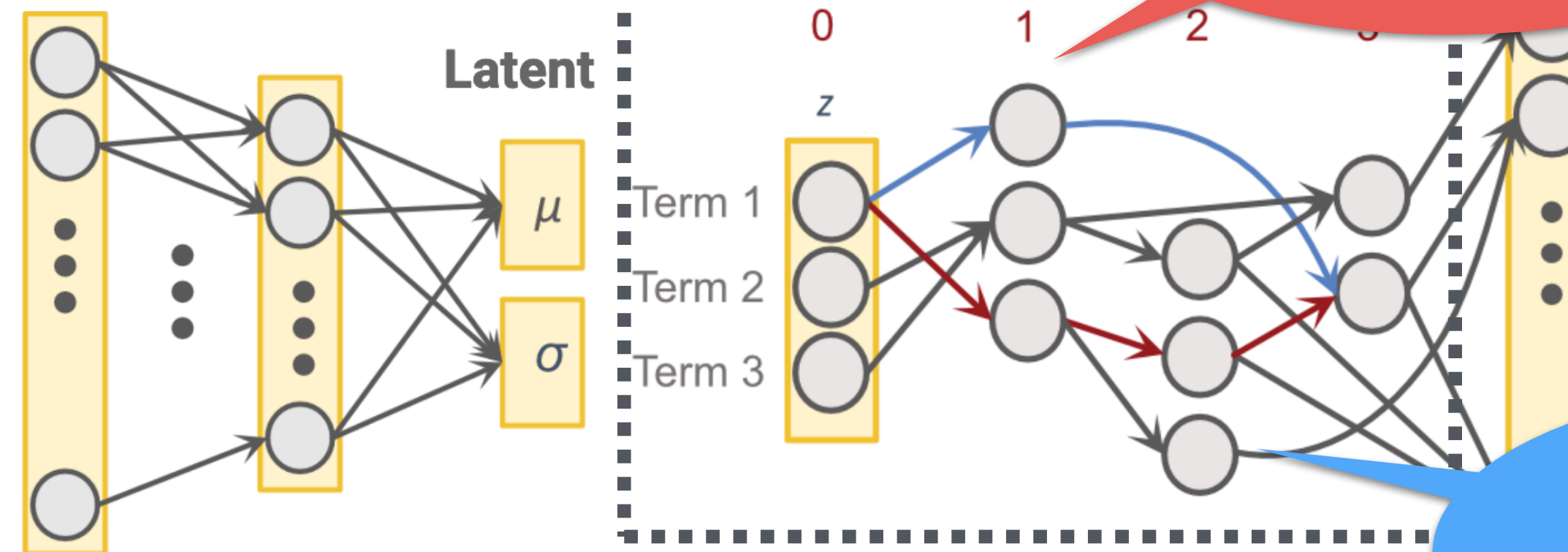
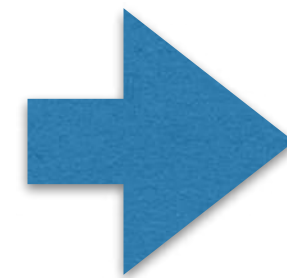
- Each decoder **node** corresponds to a **biological term** of the ontology
- Node **activation** = biological **activity**

[Doncevic et al. (2023)]

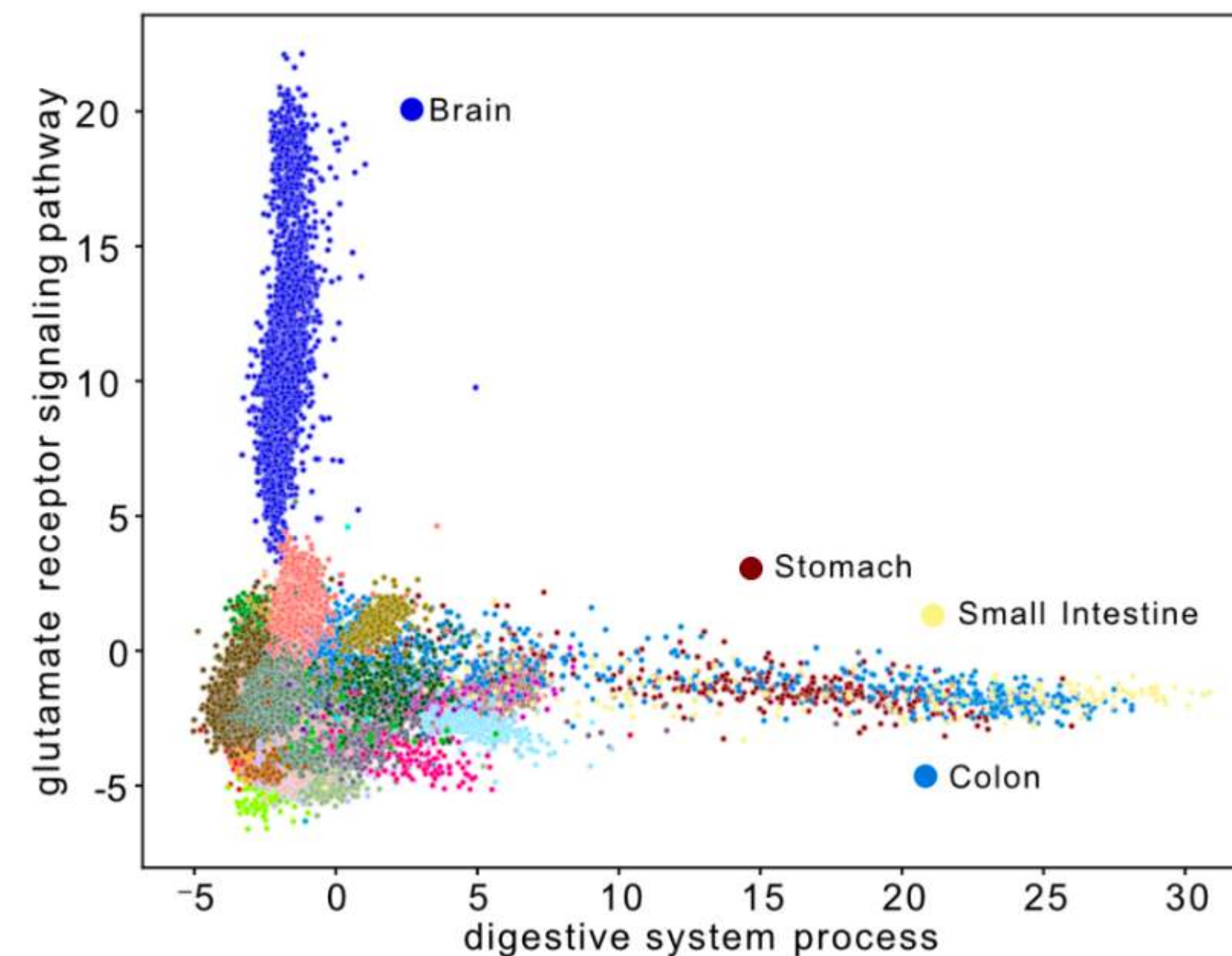
Tissue expression data



Tissue
Expression
dataset



Digestive system
process

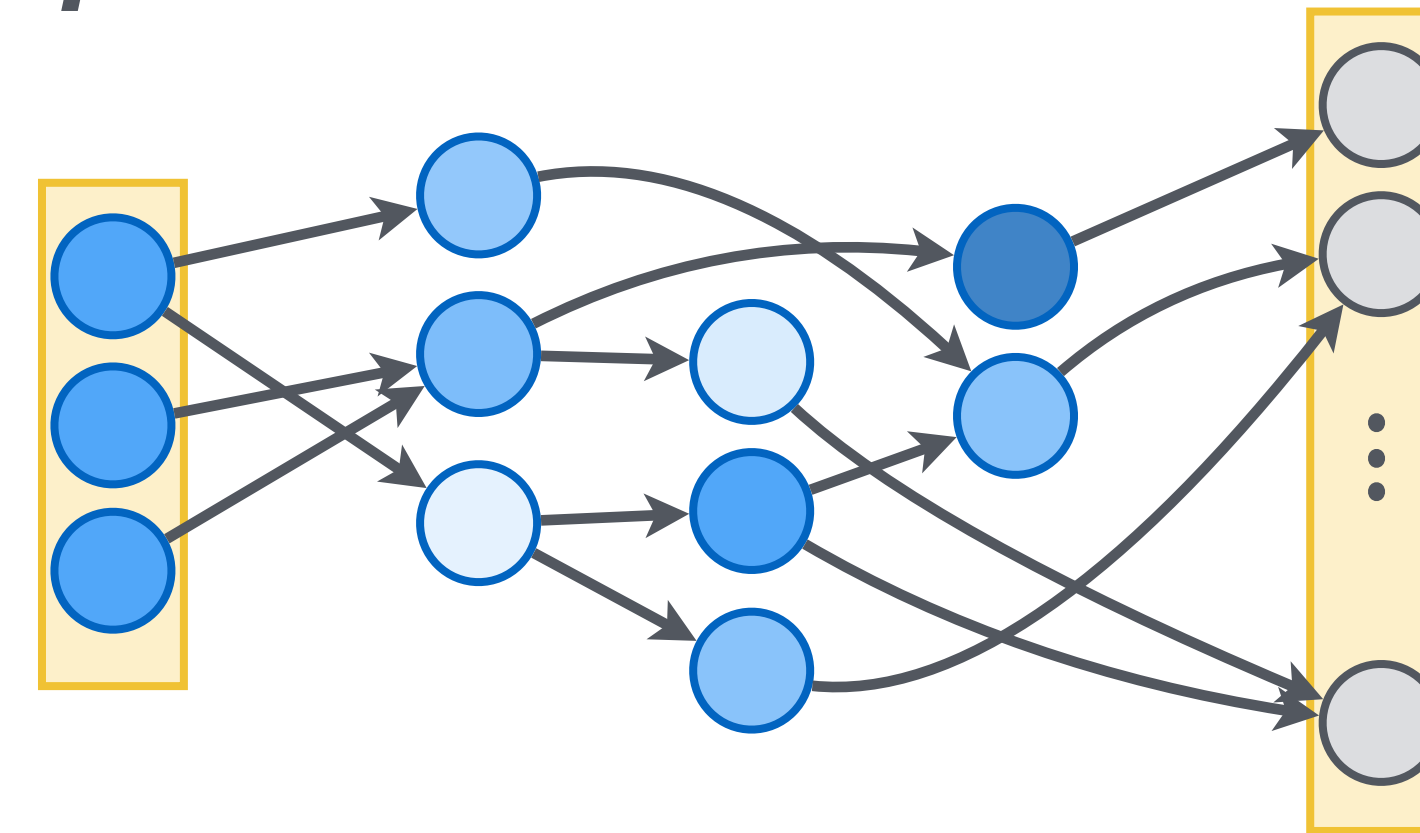
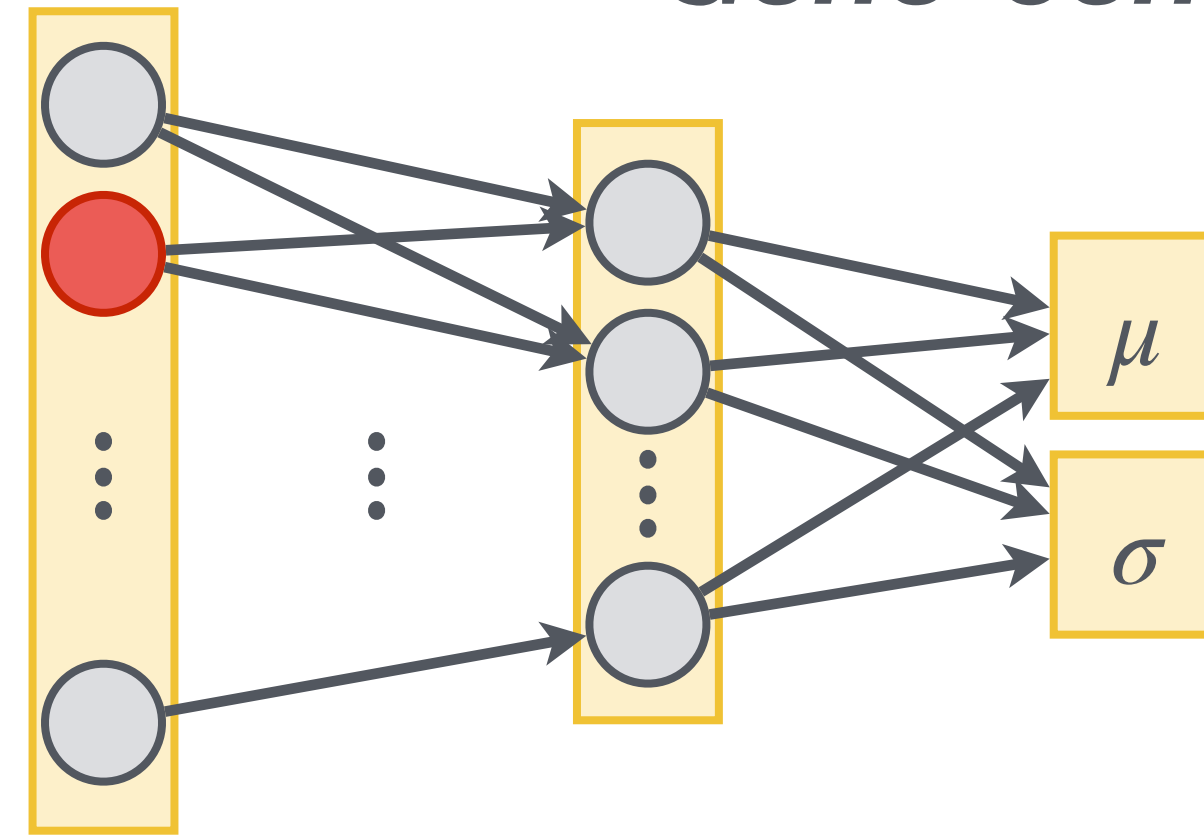


[Doncevic et al. (2023)]

Modeling perturbations

Gene-centric perturbation

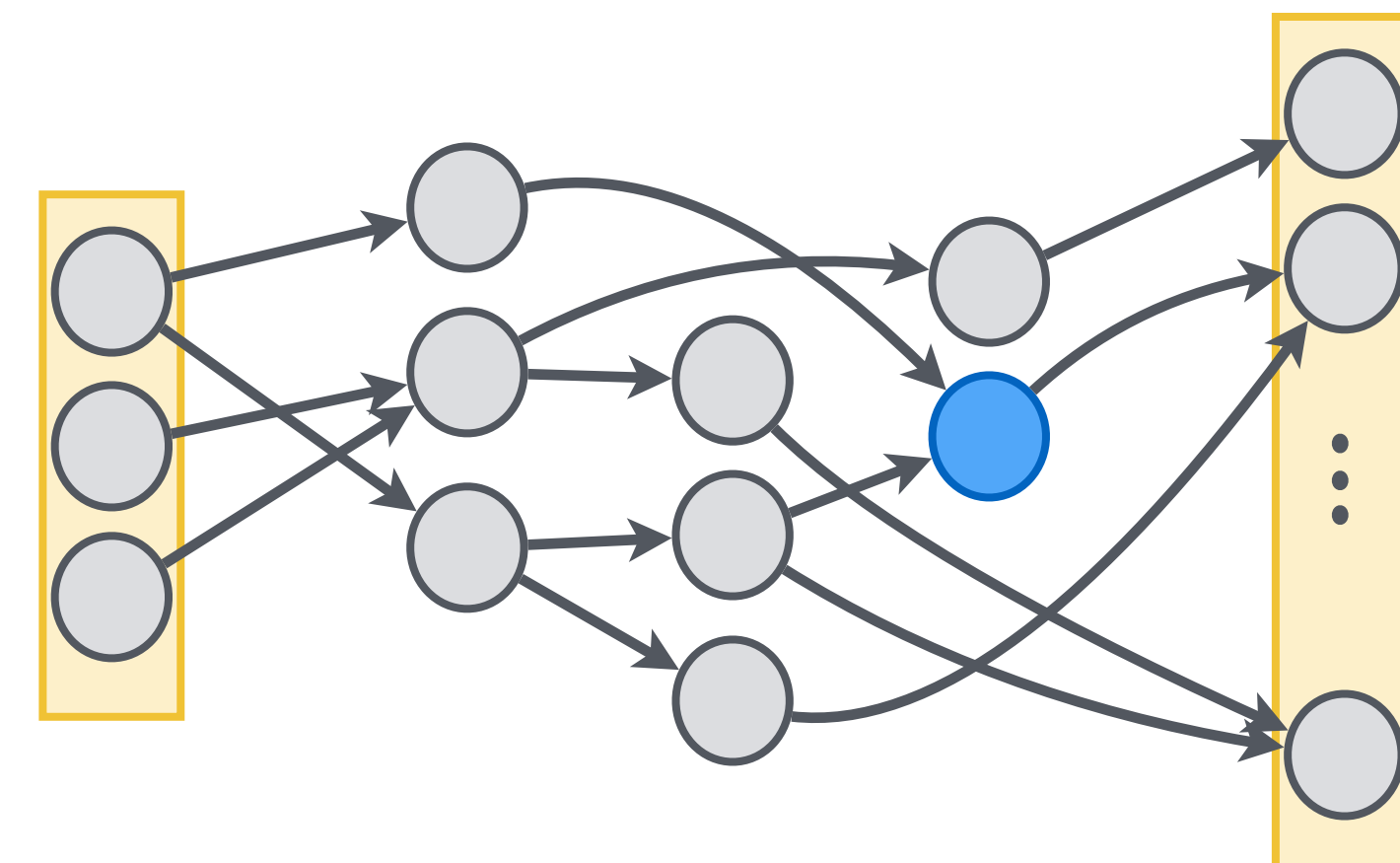
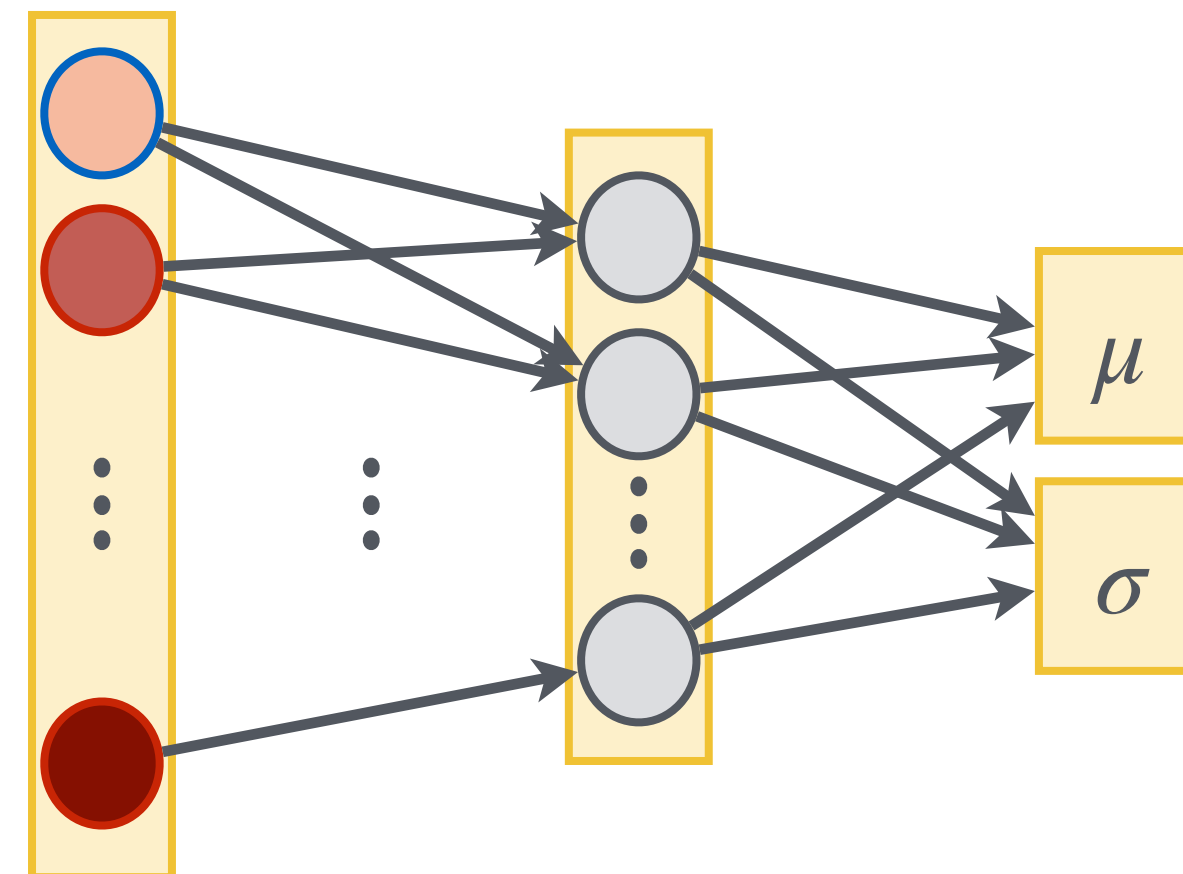
Perturb specific gene



*which processes
are most affected?*

Process-centric perturbation

*which genes have
the greatest impact?*

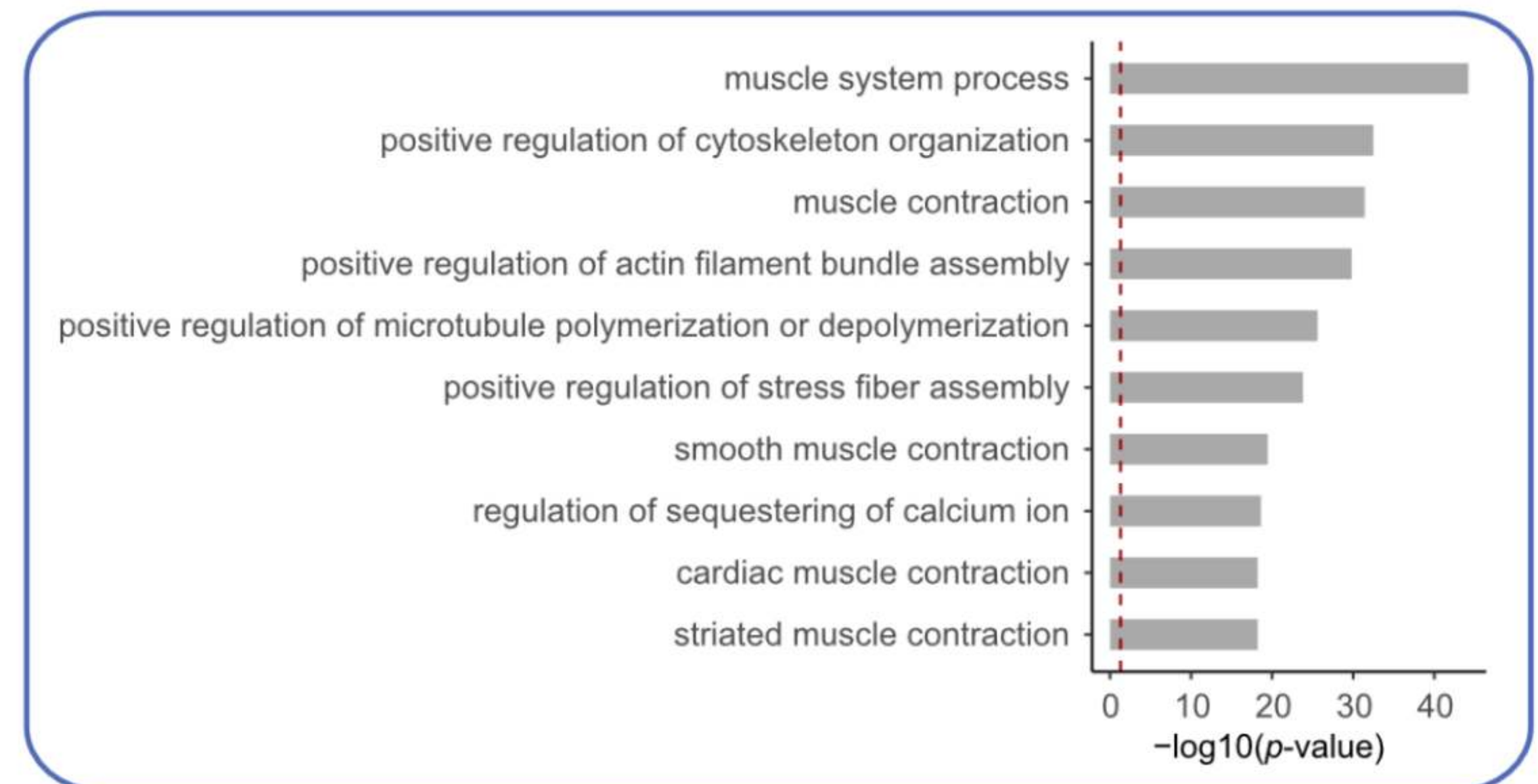
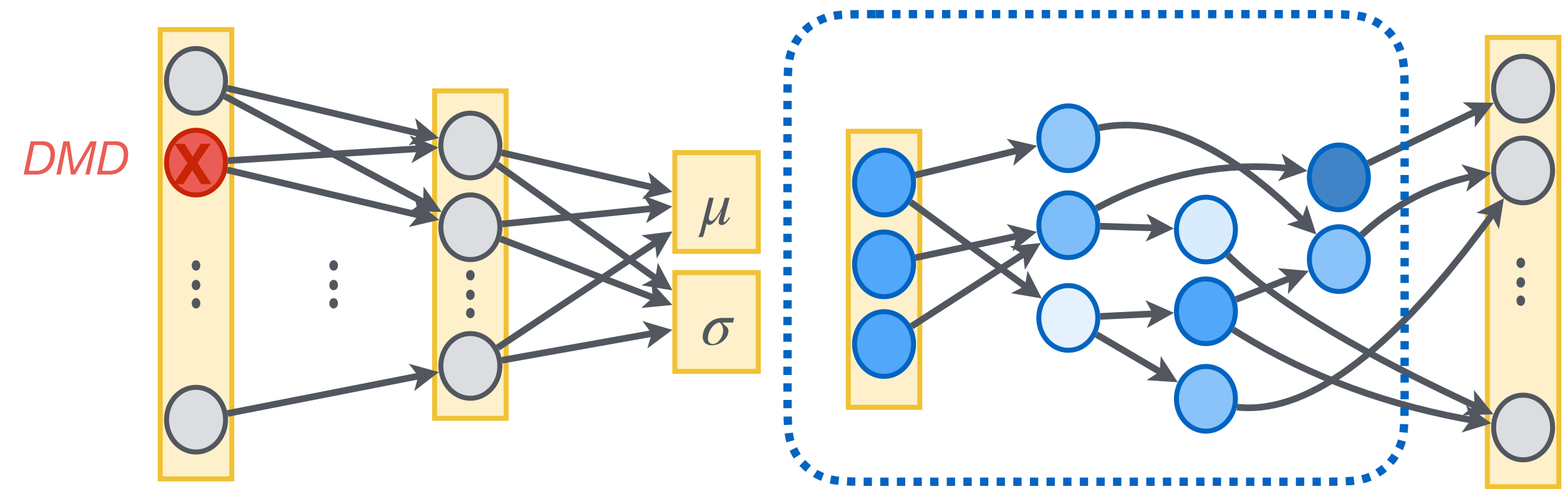


Gene-centric: genetic disease

- **Duchenne muscular dystrophy** is a genetic disorder leading to progressive muscle degeneration
- Monogenic disorder due to alterations in the dystrophin (*DMD*) gene
- **Simulate knockout** of *DMD* gene by comparing WT to in-silico knockout (in muscle samples)

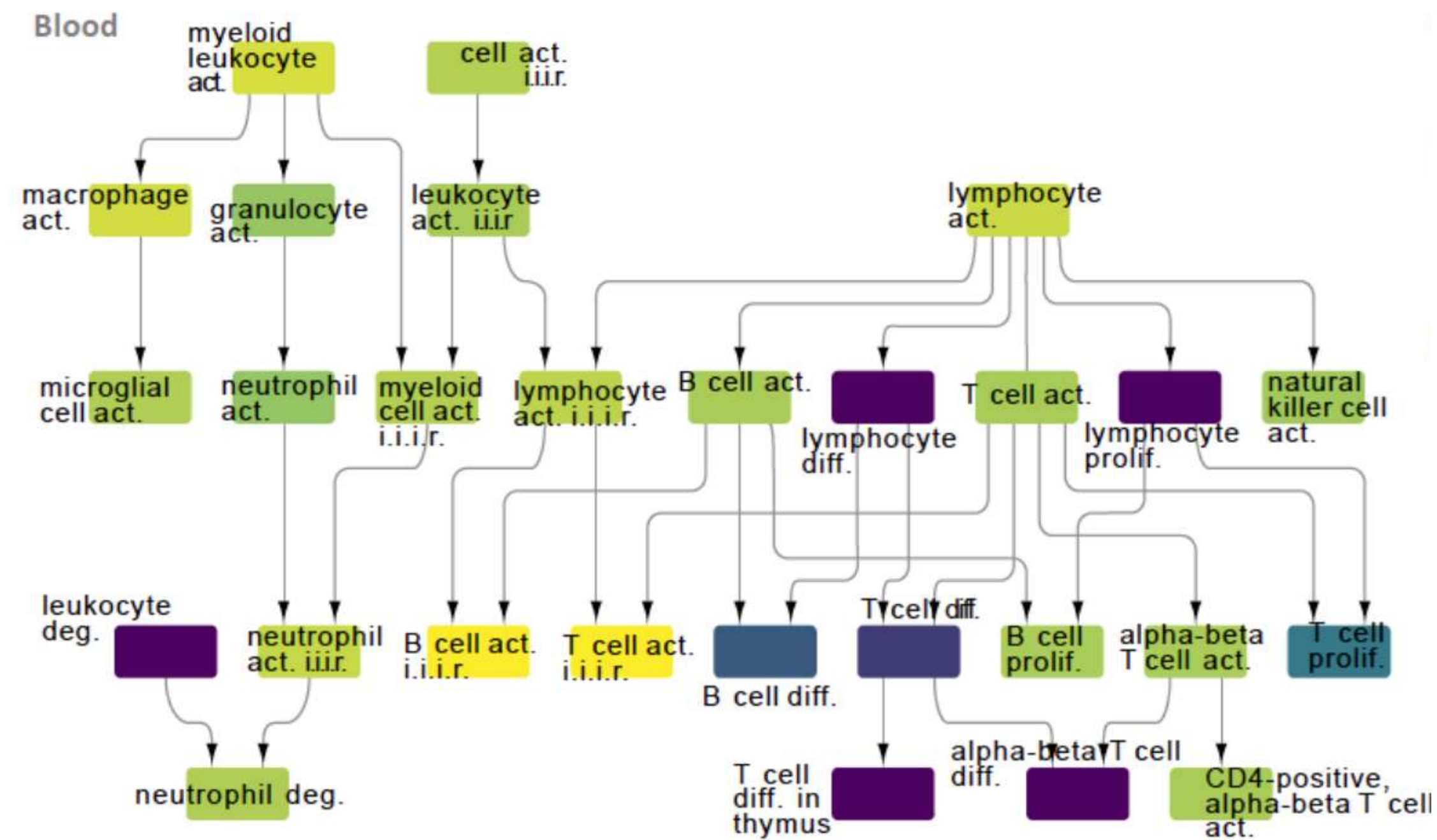
- *Model trained on all tissue samples*
- *Knockout in muscle samples*
- *Gene-ontology terms*

*Examine differential
activation of terms*

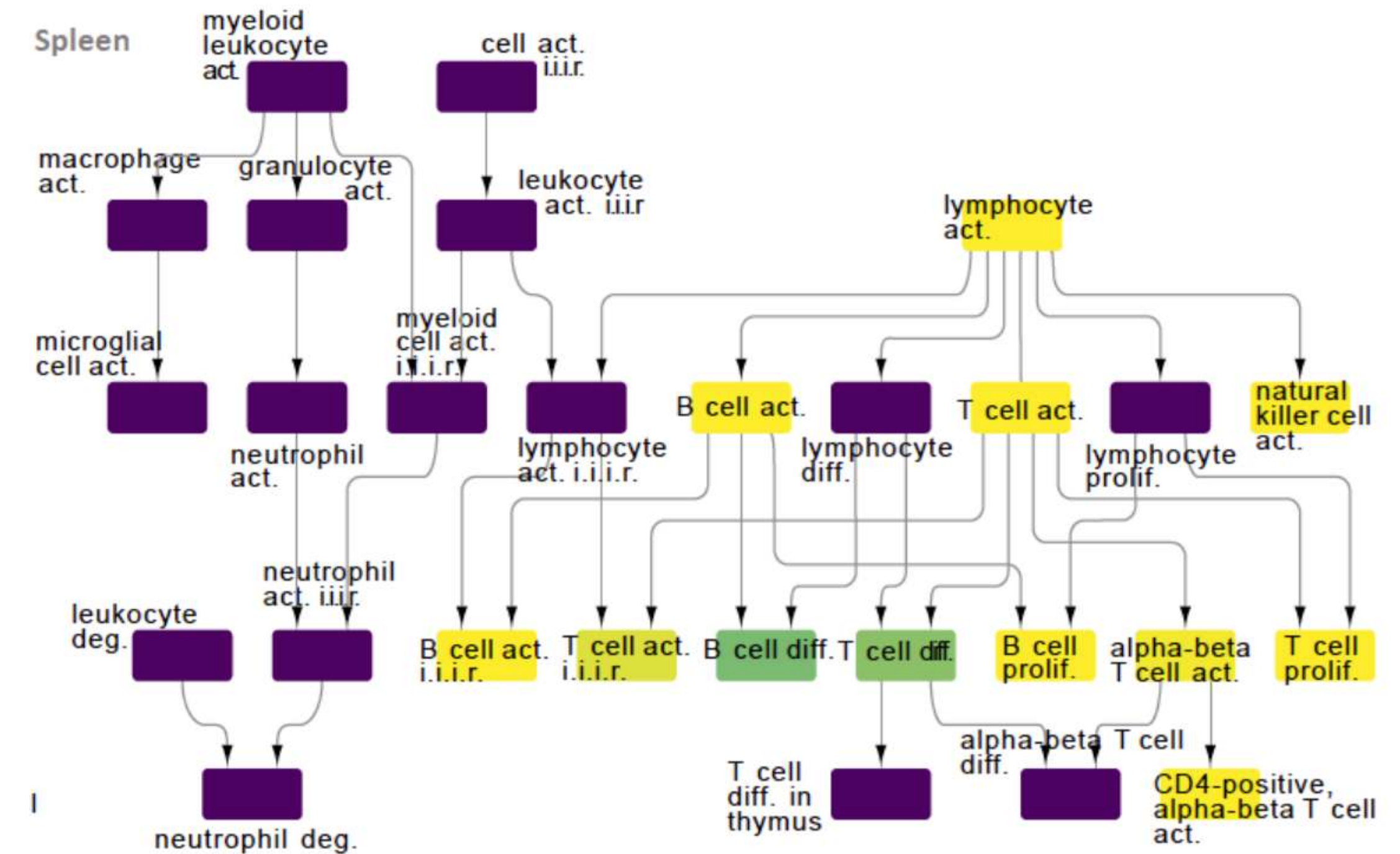


Domain specific ontology

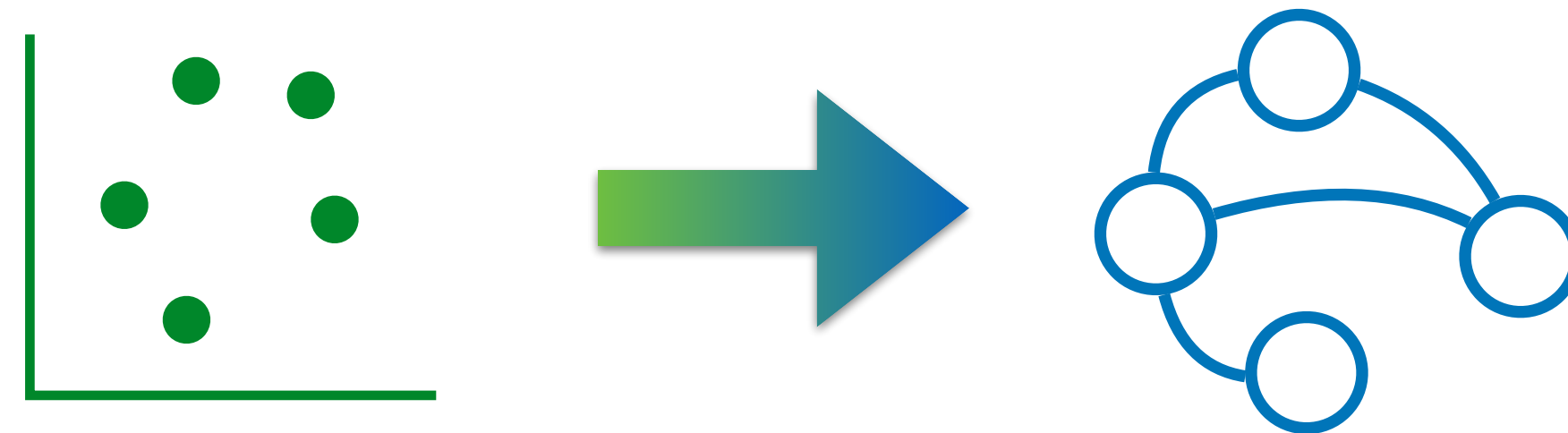
Blood



Spleen



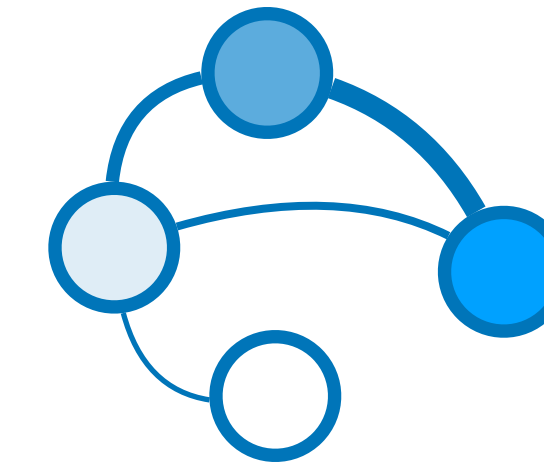
Data driven knowledge adaptation : Data to knowledge



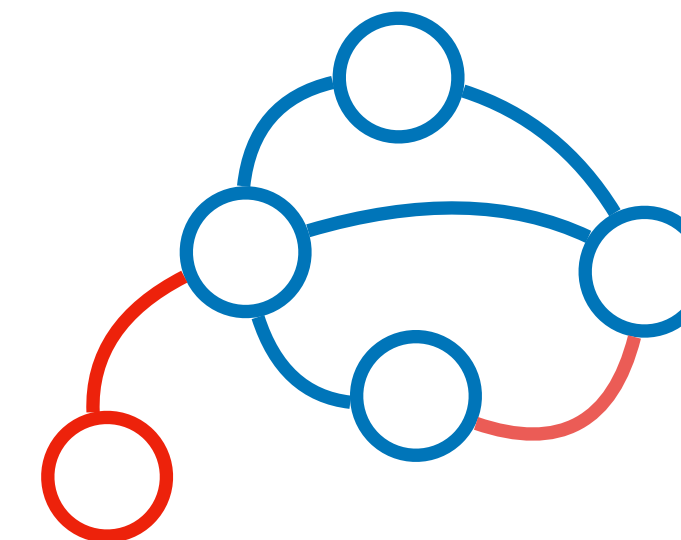
Data driven knowledge adaptation

- Current annotation resources have been mostly curated in a **context agnostic manner**
 - cell type
 - tissue
 - genetic background
 - sex
 - ...
- Define...
 - context specific **activity** (e.g. TFs, pathways)
 - context specific **components** (which genes are involved in a certain pathway?)
 - context specific **concepts**

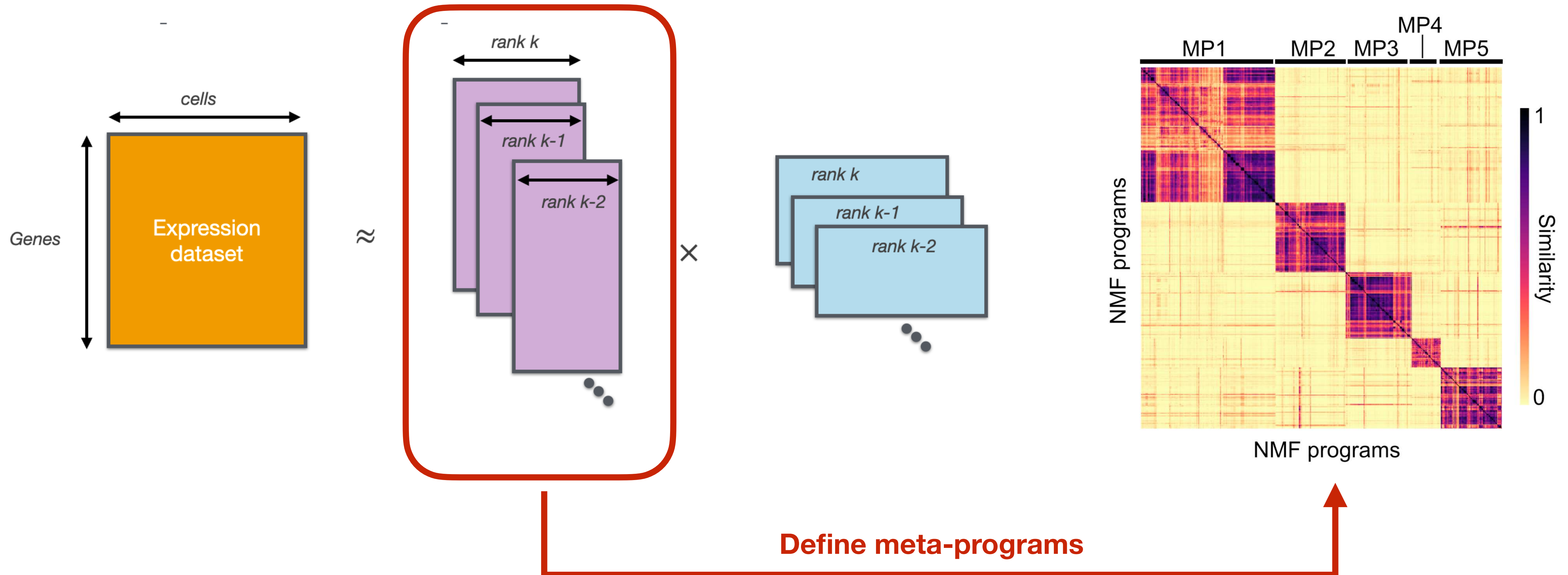
**Contextualised
knowledge**



**Knowledge
expansion**

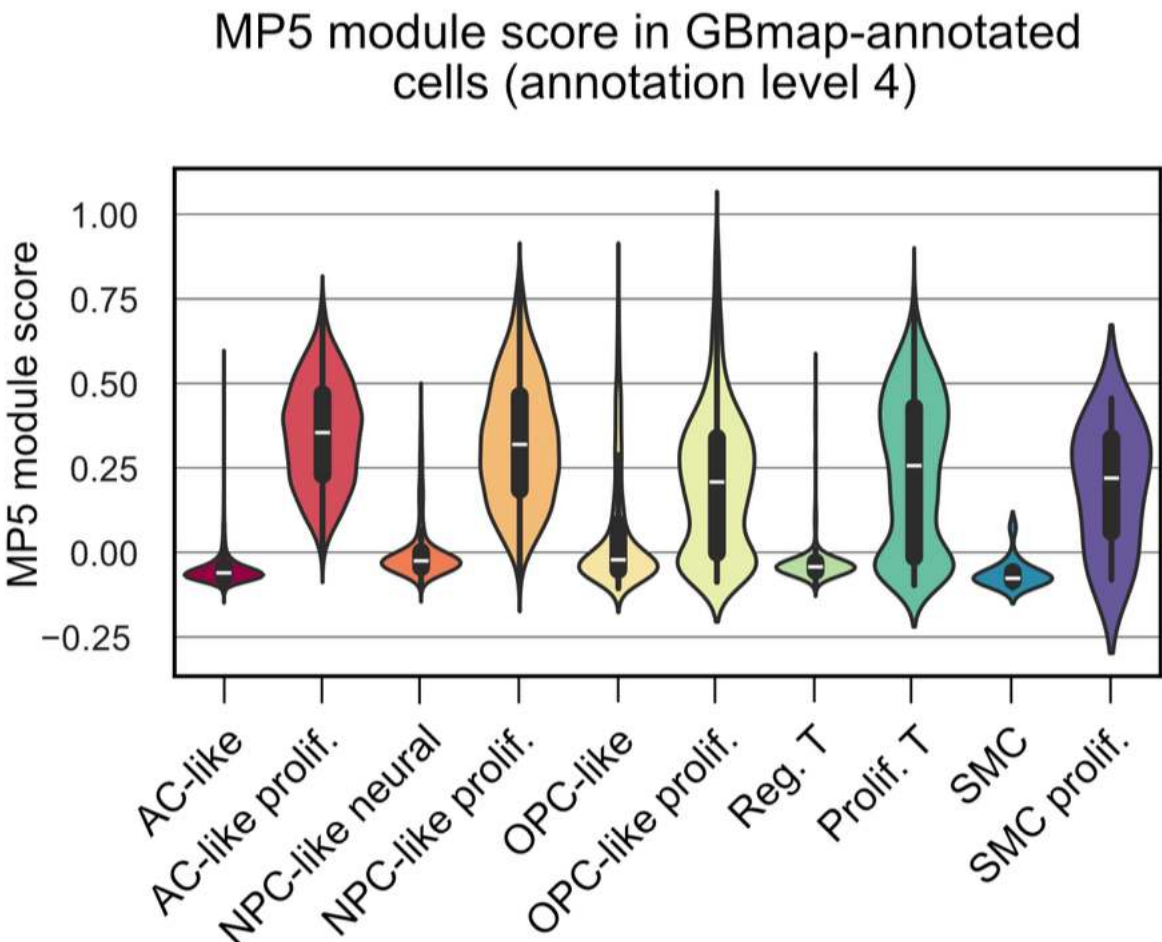
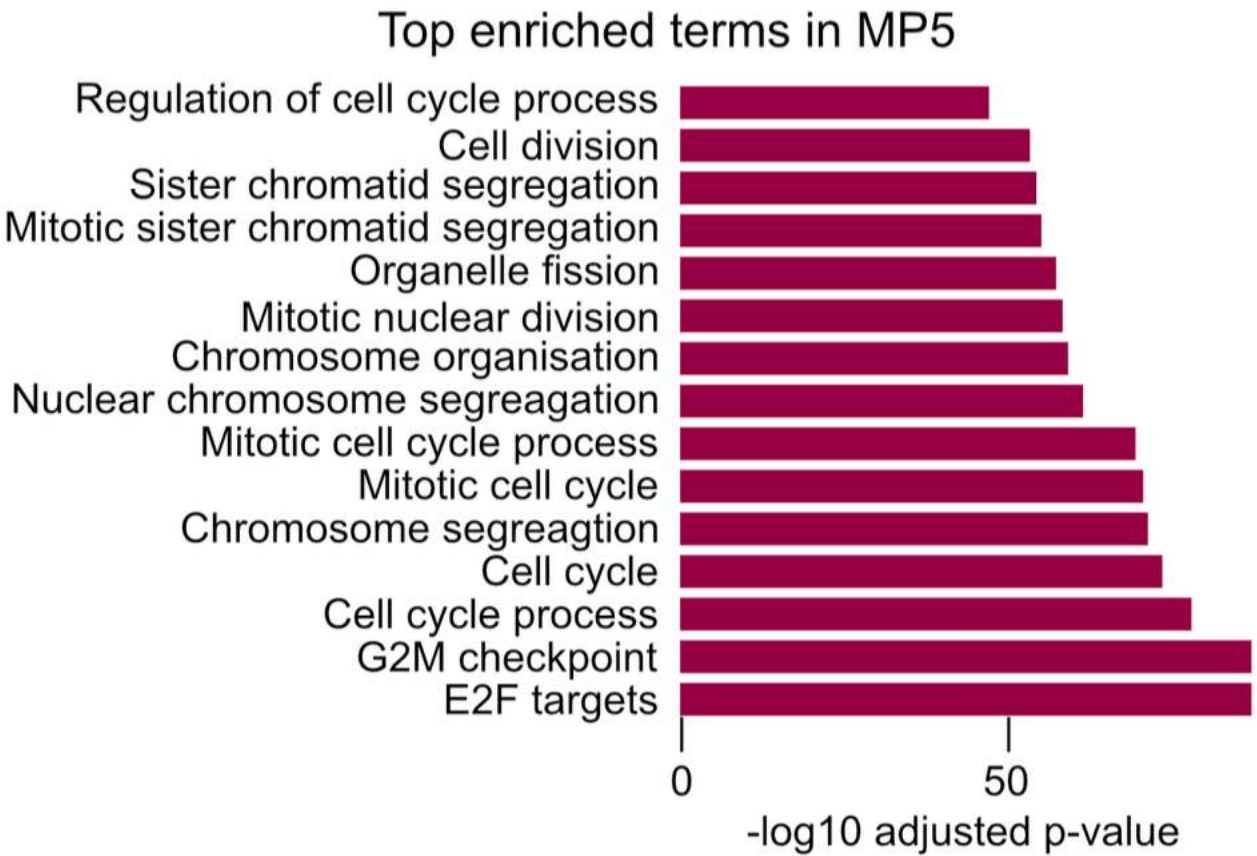
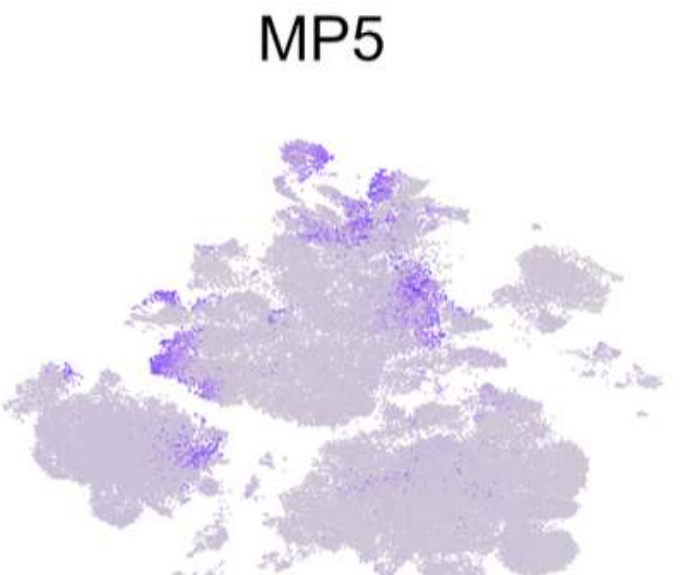
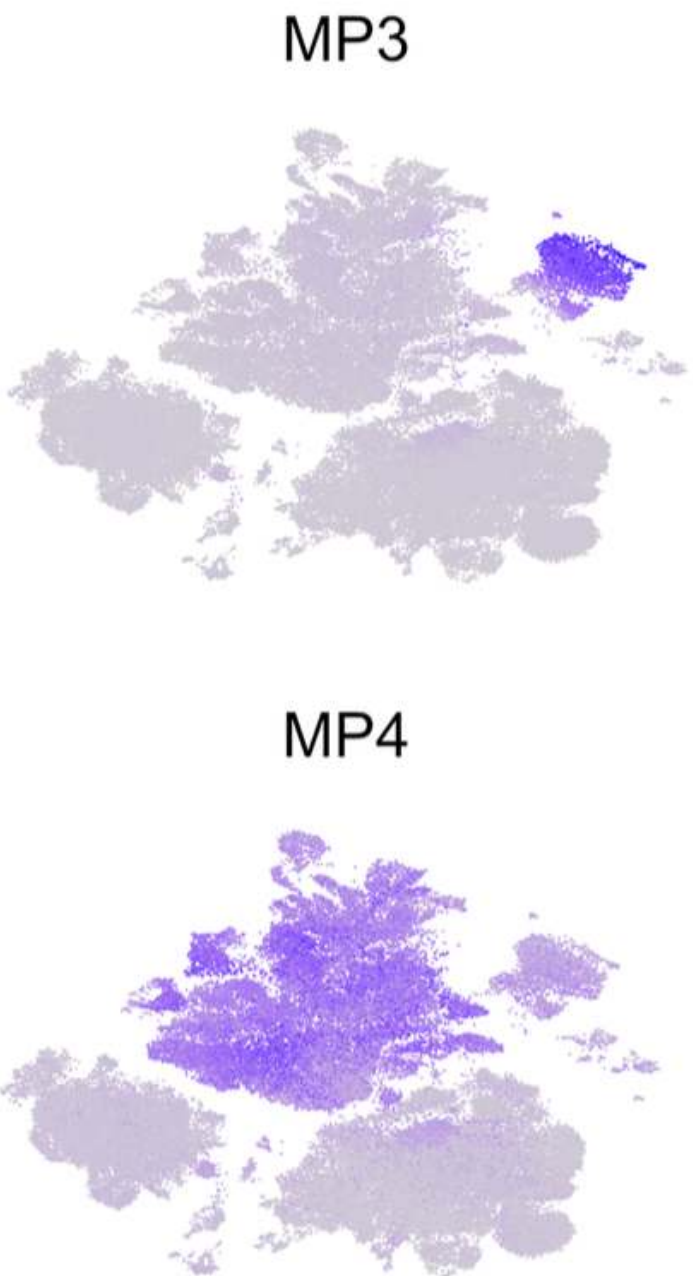
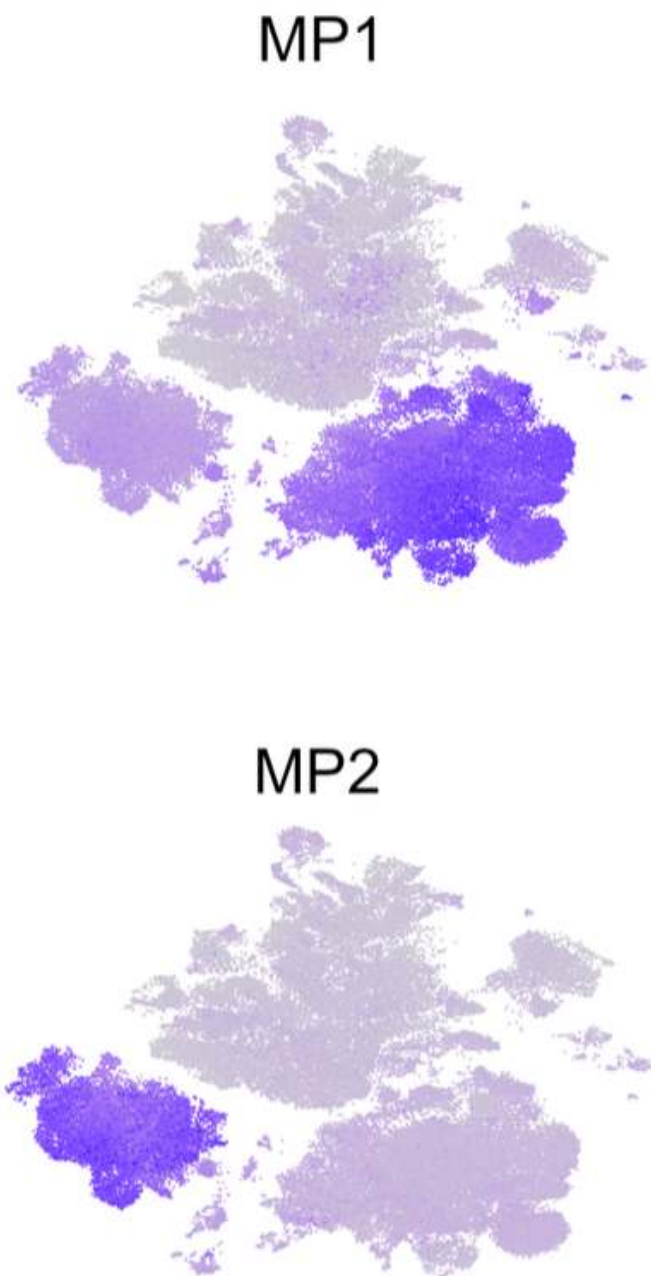
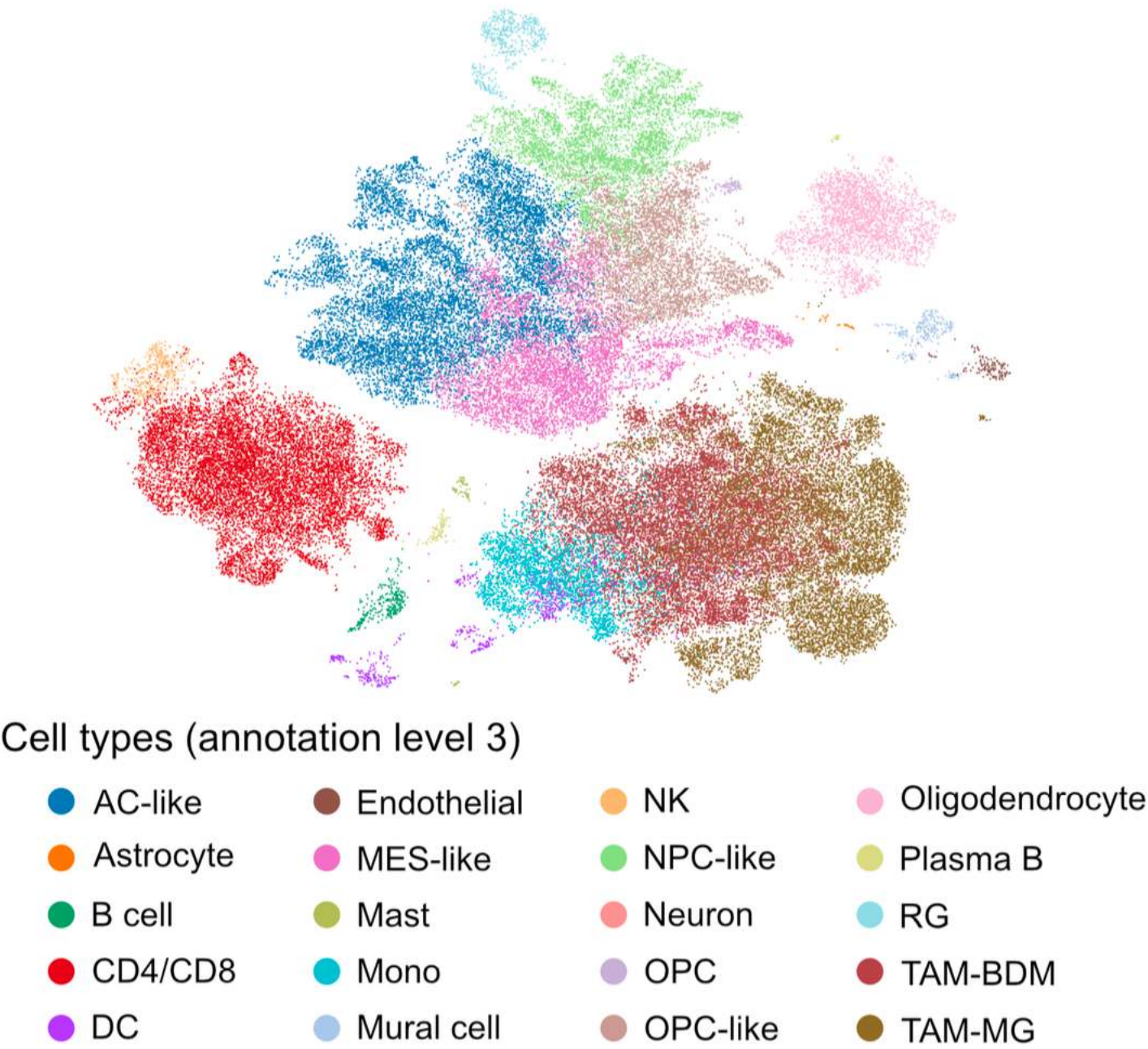


Domain specific concepts



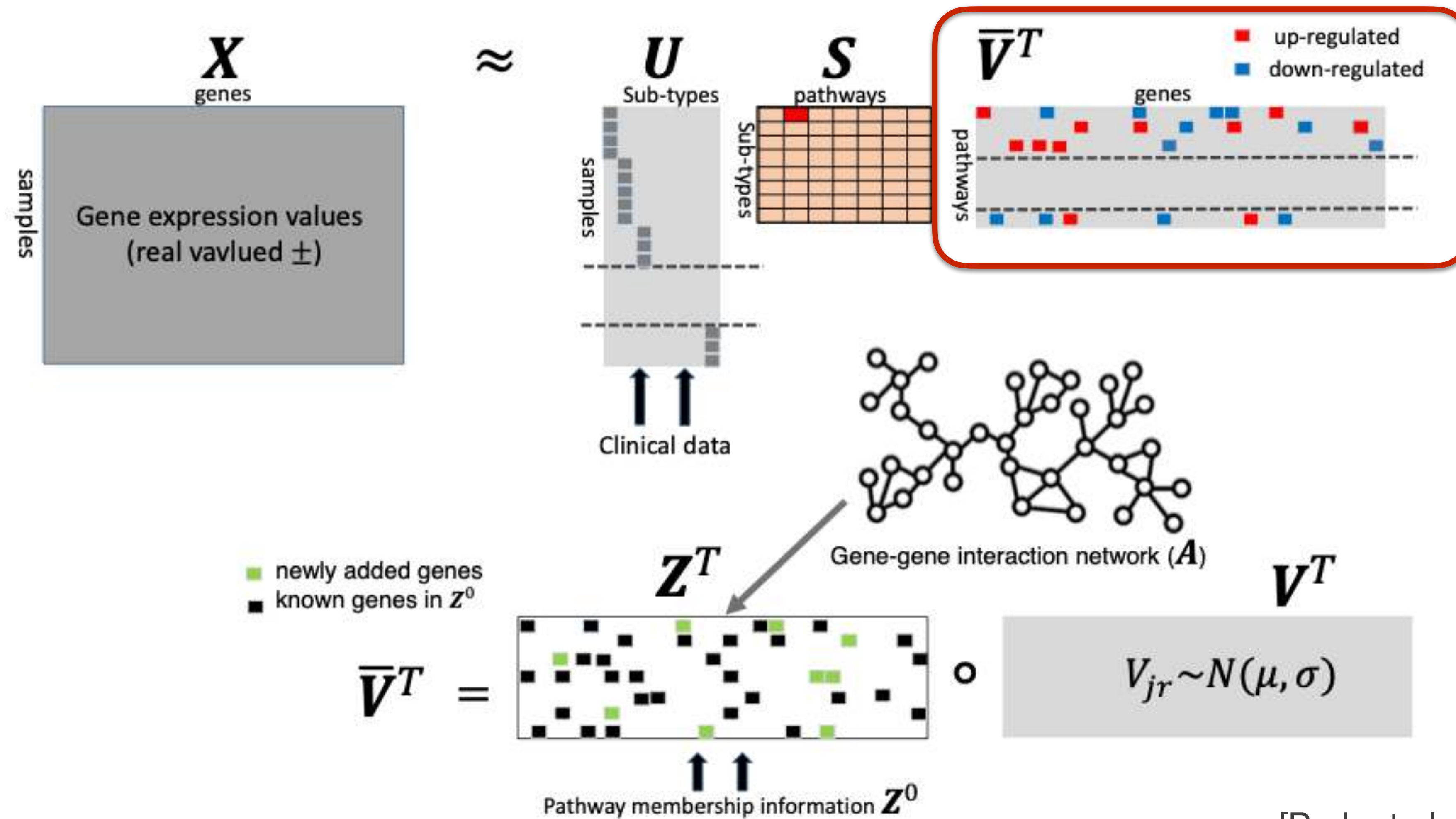
[Radig et al, 2026]

Application to Glioblastoma



[Radig et al, 2026]

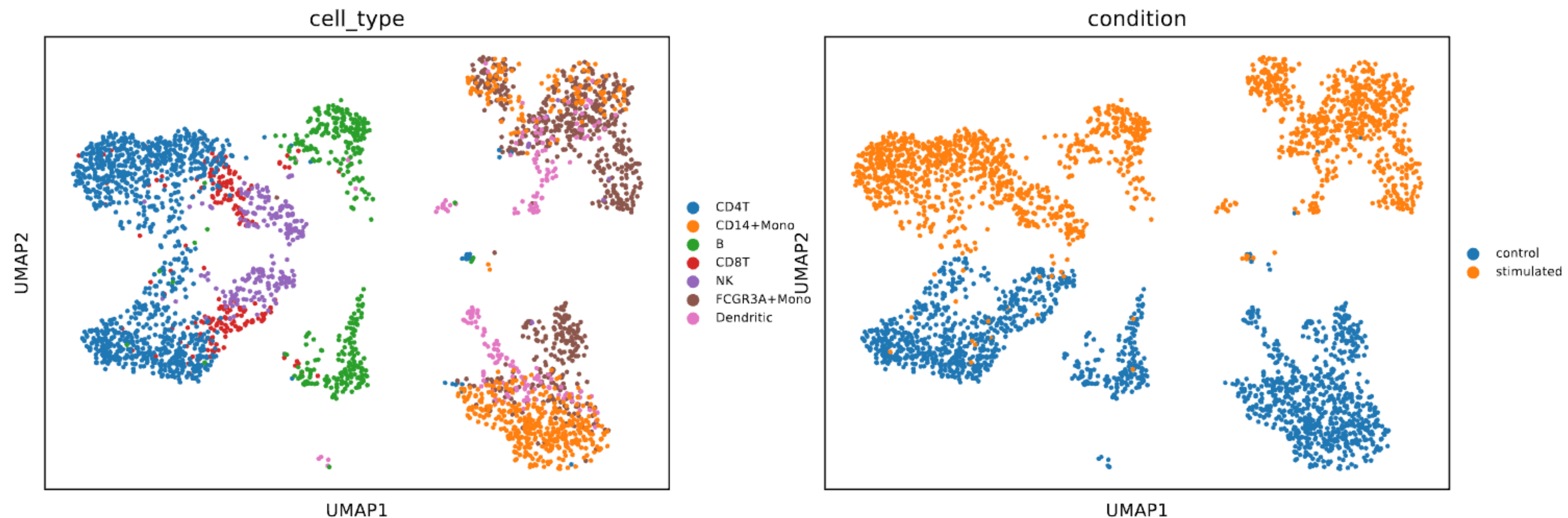
Domain specific components



[Park et al., 2019]

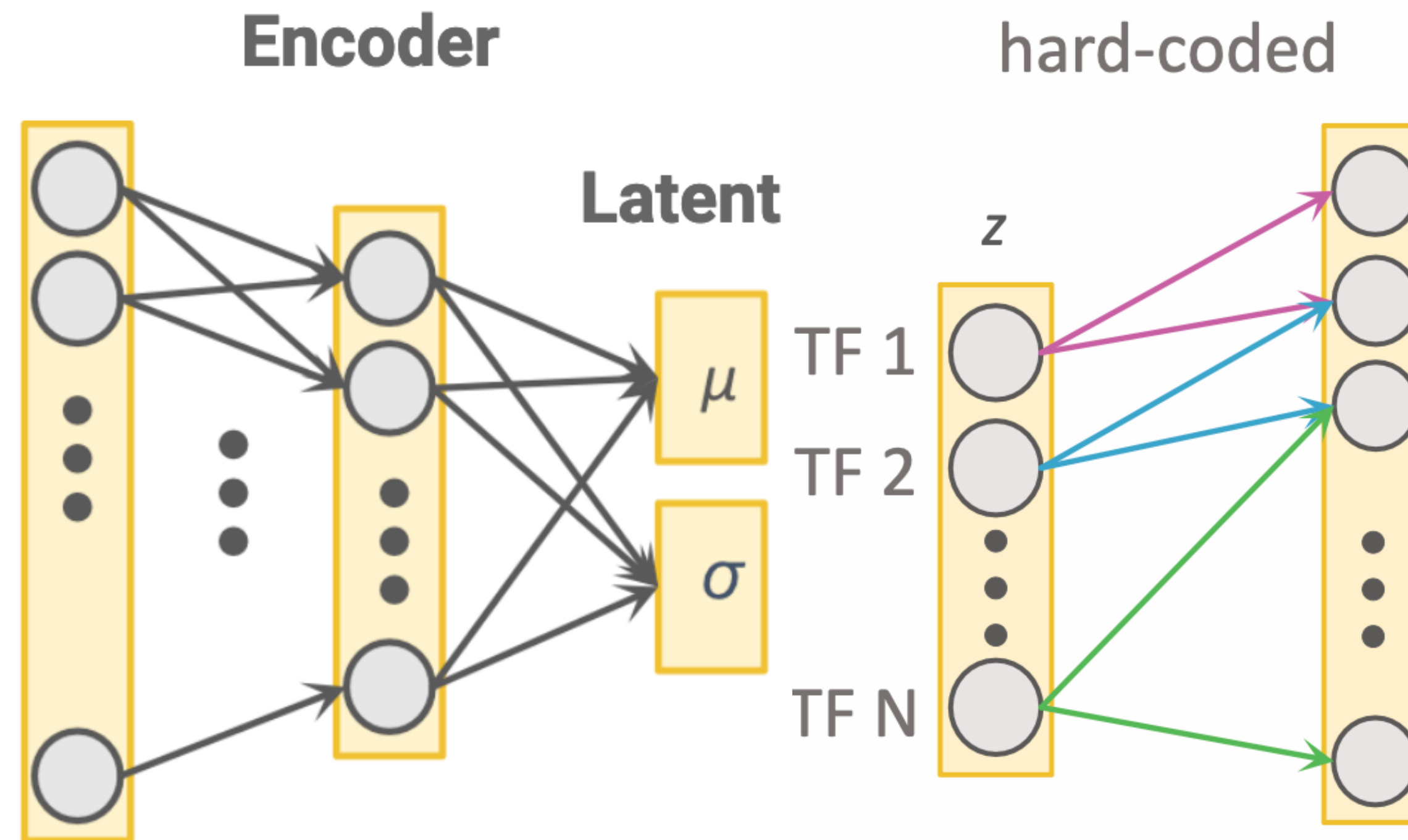
Domain specific GRNs

- Is the gene regulatory network (GRN) of a transcription factor condition-dependent?
- Can it gain / loose target genes?
- Test case: PBMC cells untreated / treated with interferon-beta
- ***How does interferon treatment reshape the gene regulatory network?***



[Anna von Bachmann]

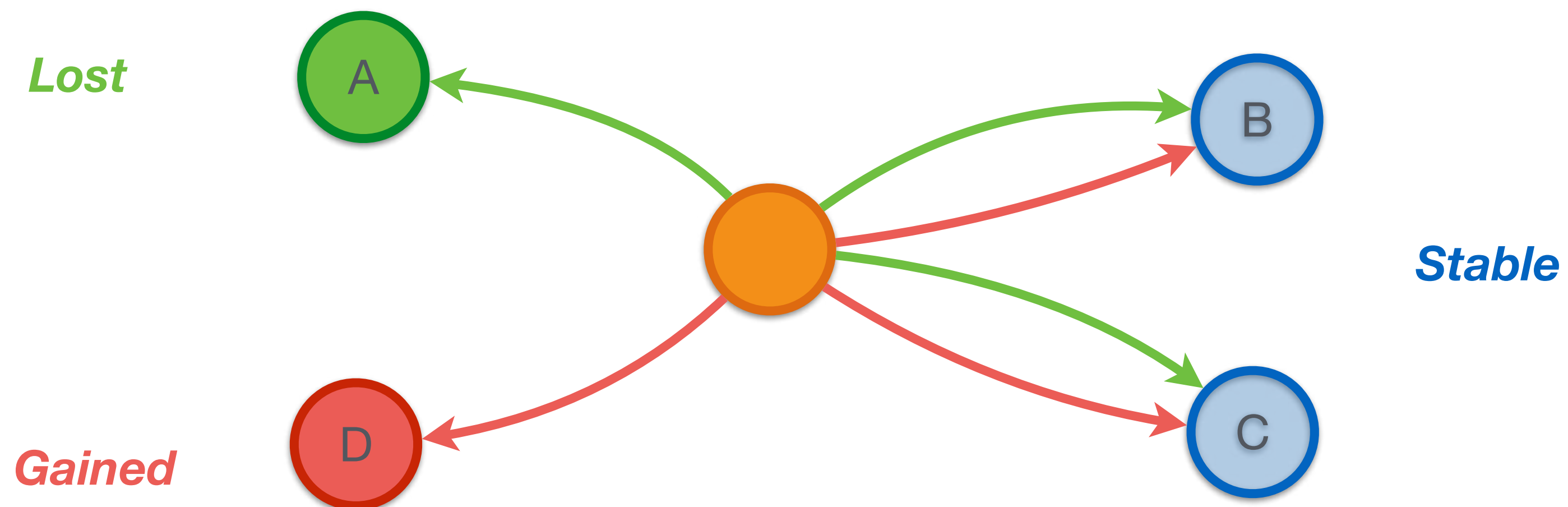
Gene regulatory network (GRN)



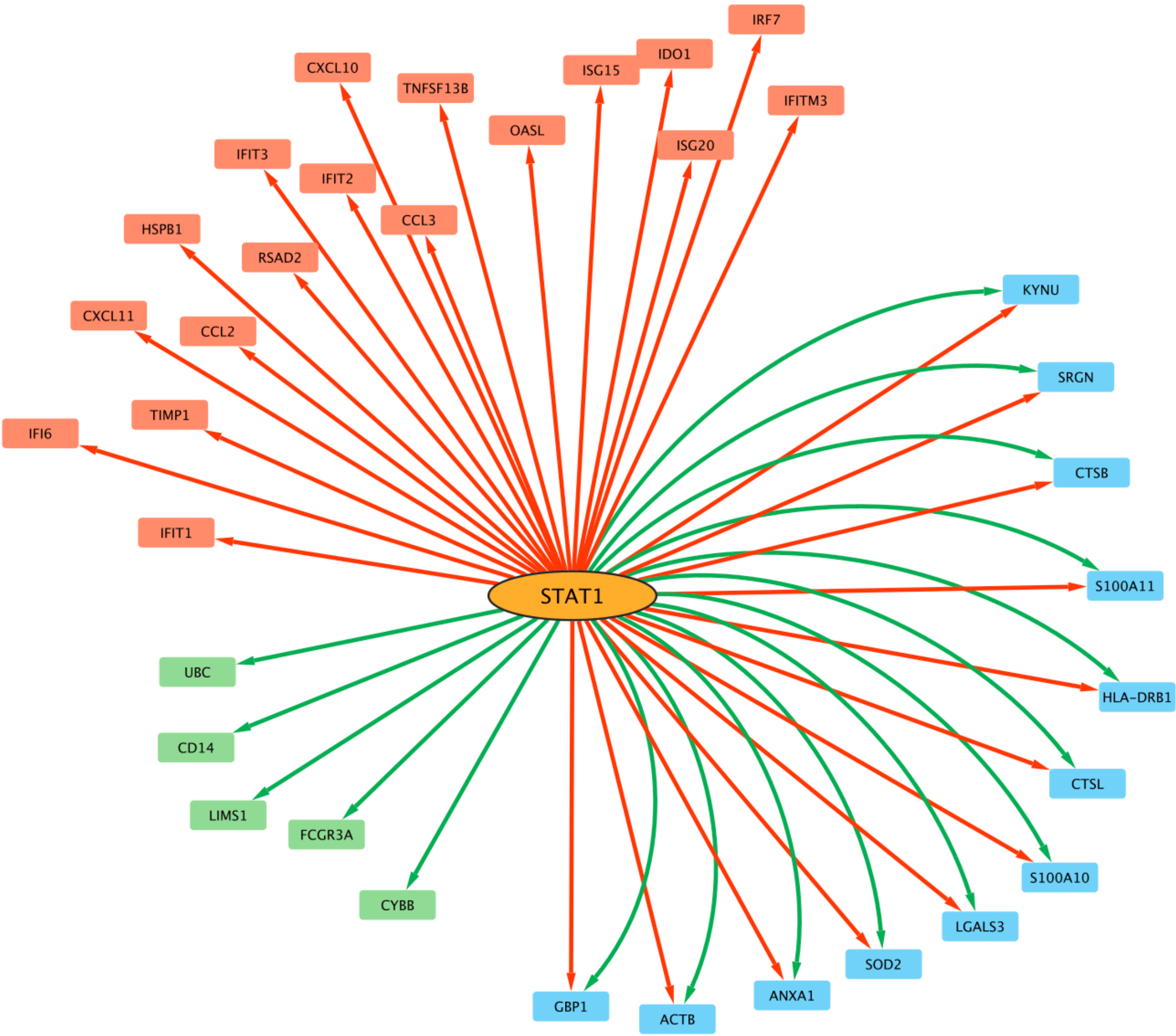
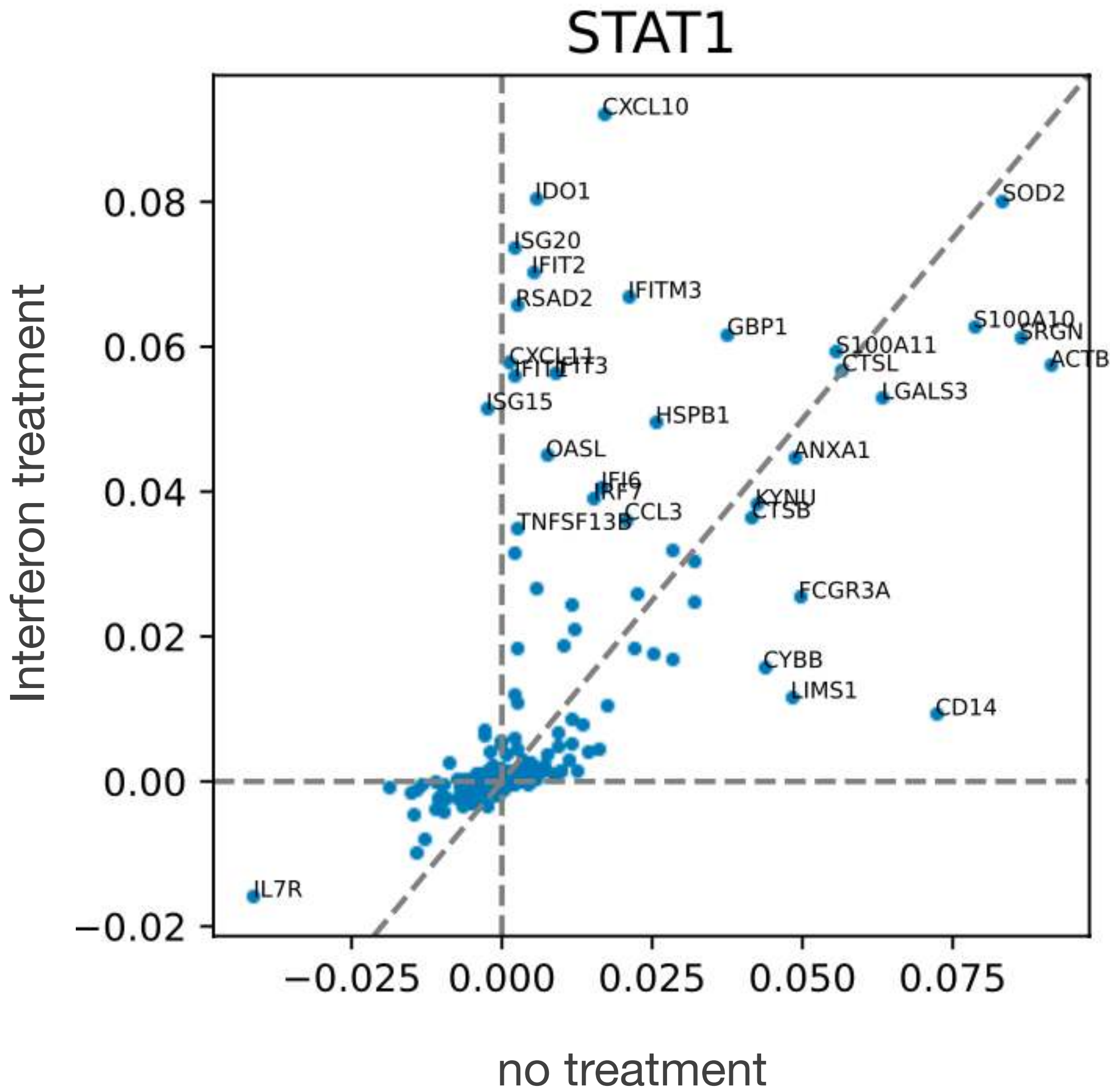
- One-layer linear decoder
- Latent space: **transcription factors**
- Decoder structure: **GRN** learned from prior knowledge or data

[Seninge et al. (2021)]

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Transcription factor rewiring

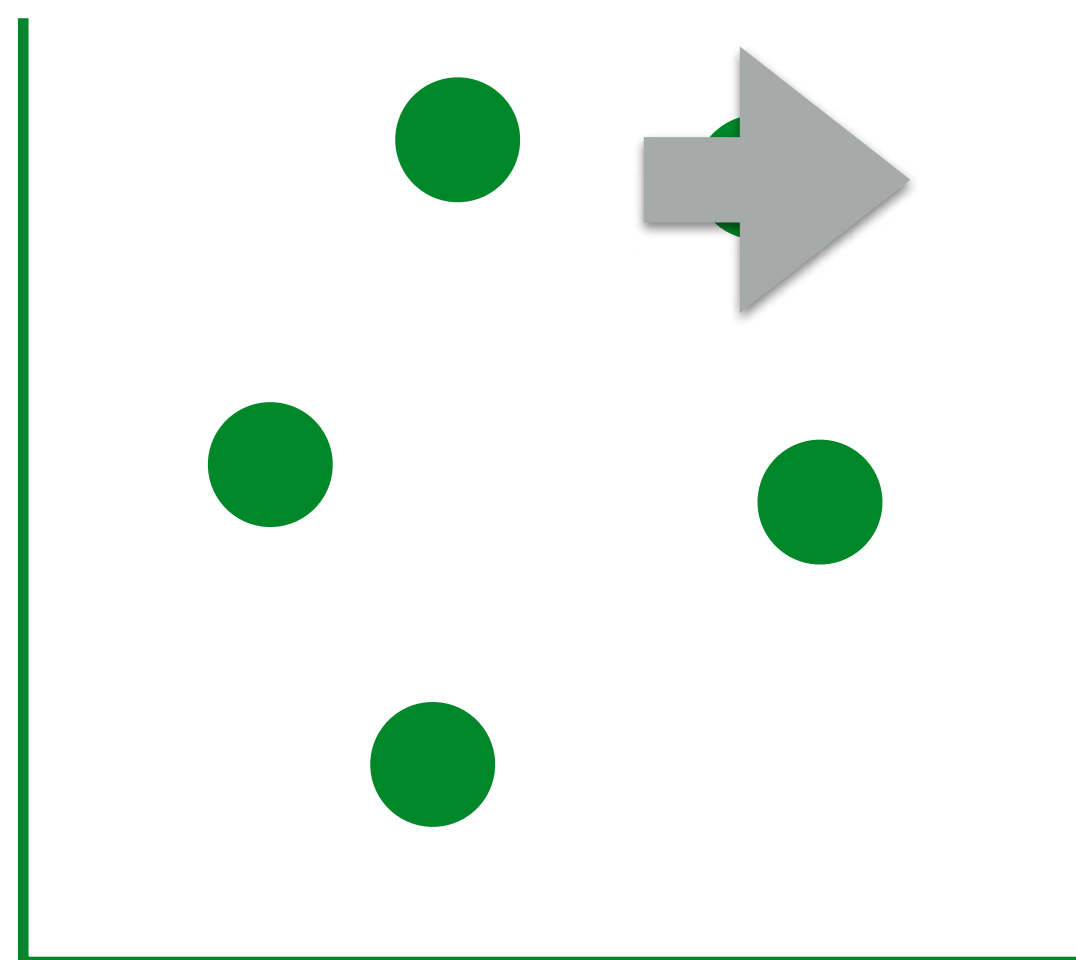


CHRIS ANDERSON

SCIENCE JUN 23, 2008 12:00 PM

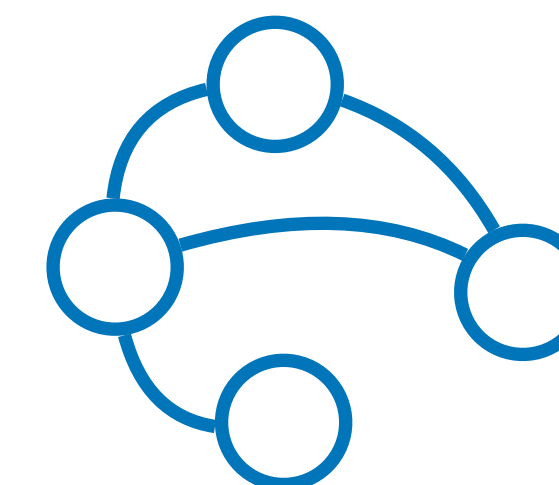
The End of Theory: The Data Deluge Makes the Scientific Method Obsolete

Data driven



**Data informed
domain knowledge**

Hypothesis driven



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